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Physical layer security

Abstract

Systems and methods of performing physical layer security are presented. In one exemplary embodiment, an electronic device includes a set of amplifier circuitry operable to output a set of output signals that collectively represents an amplified sample of an input signal having information. Further, the set of output signals corresponds to a set of bandpass components that is secured by a certain physical layer security scheme and is associated with a quantized polar representation of the input signal sample.

Inventors:	Carvalho; Paulo (Lisbon, PT), Dinis; Rui (Costa da Caparica, PT)
Applicant:	QDAComm LLC (Raleigh, NC)
Family ID:	1000008508192
Assignee:	QDAComm LLC (Raleigh, NC)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3777275	12/1972	Cox	N/A	N/A
3927379	12/1974	Cox et al.	N/A	N/A
3995227	12/1975	Lefevre	330/86	H03M 1/16
5304943	12/1993	Koontz	N/A	N/A
5850162	12/1997	Danielsons	330/149	H03F 1/3241
5872481	12/1998	Sevic et al.	N/A	N/A
6055418	12/1999	Harris	455/67.11	H01Q 3/267
6208247	12/2000	Agre et al.	N/A	N/A
6320914	12/2000	Dent	N/A	N/A
6359504	12/2001	Cozzarelli	N/A	N/A
6470055	12/2001	Feher	N/A	N/A
7085330	12/2005	Shirali	375/296	H03F 3/24
7260368	12/2006	Blumer	N/A	N/A
7272408	12/2006	Dalal et al.	N/A	N/A
7362821	12/2007	Shirali	375/285	H03F 1/3247
7480395	12/2008	Parunak et al.	N/A	N/A
7526261	12/2008	Sorrells et al.	N/A	N/A
7629843	12/2008	Paul et al.	N/A	N/A
7715811	12/2009	Kenington	N/A	N/A
7729675	12/2009	Krone	N/A	N/A
7737786	12/2009	Plaze et al.	N/A	N/A
7889811	12/2010	Byun et al.	N/A	N/A
7937049	12/2010	Phillips et al.	N/A	N/A
7941157	12/2010	Srinivasan et al.	N/A	N/A
7979574	12/2010	Gillo et al.	N/A	N/A
8289084	12/2011	Morimoto et al.	N/A	N/A
8299266	12/2011	Parlanti et al.	N/A	N/A
8362832	12/2012	Kim et al.	N/A	N/A
8451053	12/2012	Perreault et al.	N/A	N/A
8706794	12/2013	Fleizach	708/622	G06F 7/4806
9012415	12/2014	Da Silva Correia et al.	N/A	N/A
9020453	12/2014	Briffa et al.	N/A	N/A
9166528	12/2014	Sorrells et al.	N/A	N/A
9413303	12/2015	Mostov et al.	N/A	N/A
9438172	12/2015	Cohen	N/A	N/A
9537456	12/2016	Briffa et al.	N/A	N/A
9712114	12/2016	Drogi	N/A	H03F 1/0227
9749786	12/2016	Pandey et al.	N/A	N/A
10069467	12/2017	Carvalho	N/A	H03F 3/195
10338193	12/2018	Beko et al.	N/A	N/A
10511359	12/2018	Carvalho et al.	N/A	N/A
10597316	12/2019	Gomes De Oliveira et al.	N/A	N/A
11095253	12/2020	Carvalho et al.	N/A	N/A

11211900	12/2020	Carvalho et al.	N/A	N/A
11265200	12/2021	Carvalho et al.	N/A	N/A
12308871	12/2024	Chavez	N/A	H04B 1/1027
2001/0016013	12/2000	Feher	N/A	N/A
2002/0093641	12/2001	Ortyn	356/28	G01N 15/147
2003/0174018	12/2002	Cooper et al.	N/A	N/A
2003/0215023	12/2002	Chang et al.	N/A	N/A
2004/0202255	12/2003	Dallal	N/A	N/A
2005/0032483	12/2004	Klomsdorf et al.	N/A	N/A
2005/0069067	12/2004	Zerbe	375/353	H04L 25/4917
2006/0006945	12/2005	Burns et al.	N/A	N/A
2006/0158254	12/2005	Johnson	330/149	H03F 1/3247
2007/0210835	12/2006	Aghtar	327/94	G11C 27/026
2007/0223622	12/2006	Bang et al.	N/A	N/A
2007/0293237	12/2006	Correal et al.	N/A	N/A
2009/0311980	12/2008	Sjoland	N/A	N/A
2010/0265799	12/2009	Cevher et al.	N/A	N/A
2010/0283652	12/2009	Nadiguebe	330/252	H03F 3/45475
2011/0038440	12/2010	Balbach et al.	N/A	N/A
2011/0187437	12/2010	Perrault et al.	N/A	N/A
2011/0212696	12/2010	Hahn	455/83	H04B 1/525
2011/0234315	12/2010	Chen	330/149	H03F 1/3241
2011/0294449	12/2010	Budianu	455/245.1	H03G 3/3052
2011/0316747	12/2010	Budianu	342/387	G01S 11/00
2012/0063544	12/2011	Dai	375/296	H03F 1/3247
2012/0326927	12/2011	Nikkels	N/A	N/A
2013/0021104	12/2012	Schmidt	N/A	N/A
2013/0301454	12/2012	Seol et al.	N/A	N/A
2014/0062793	12/2013	AlSindi et al.	N/A	N/A
2014/0350237	12/2013	da Silva Correia et al.	N/A	N/A
2015/0065039	12/2014	Nii	455/41.1	H01F 38/14
2015/0221313	12/2014	Purnhagen et al.	N/A	N/A
2015/0288437	12/2014	Raju et al.	N/A	N/A
2016/0079924	12/2015	Ogawa	330/296	H03F 1/0261
2016/0164466	12/2015	Briffa et al.	N/A	N/A
2017/0090025	12/2016	Wang	N/A	N/A
2017/0244582	12/2016	Gal	N/A	H04L 25/03885
2017/0245908	12/2016	Lizauckas	N/A	B01D 46/429
2017/0279501	12/2016	Kim et al.	N/A	N/A
2017/0280281	12/2016	Pandey et al.	N/A	N/A
2017/0346559	12/2016	Eroglu et al.	N/A	N/A
2018/0138862	12/2017	Balteanu et al.	N/A	N/A
2020/0403839	12/2019	Carvalho	N/A	H03F 3/189
2021/0006207	12/2020	Carvalho et al.	N/A	N/A
2021/0359651	12/2020	Carvalho et al.	N/A	N/A
2024/0223229	12/2023	Molina	N/A	H03F 3/24

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110710114	12/2019	CN	N/A
114402569	12/2021	CN	N/A
114450888	12/2021	CN	N/A
1469649	12/2003	EP	N/A
1611401	12/2005	EP	N/A
2552406	12/2012	EP	N/A
3306816	12/2017	EP	N/A
6242548	12/2016	JP	N/A
201742370	12/2016	TW	N/A
9611149	12/1995	WO	N/A
02075226	12/2001	WO	N/A
2004090437	12/2003	WO	N/A
2009101518	12/2008	WO	N/A
2011119059	12/2010	WO	N/A
2011119059	12/2010	WO	N/A
2015183114	12/2014	WO	N/A
2018160083	12/2017	WO	N/A
2019074384	12/2018	WO	N/A
2020076172	12/2019	WO	N/A
2020076172	12/2019	WO	N/A
2020177552	12/2019	WO	N/A
2020261176	12/2019	WO	N/A
2021069192	12/2020	WO	N/A
2021154106	12/2020	WO	N/A
2021221522	12/2020	WO	N/A
2021221523	12/2020	WO	N/A
2022108463	12/2021	WO	N/A

OTHER PUBLICATIONS

A Gusmao and N. Esteves, “ENCAP-4: an OQPSK-type modulation technique for digital radio”, IEEE Proc. I, Commun. Speech Vis., vol. 135, No. 1, pp. 105-110, Feb. 1988. cited by applicant

A. Araujo, J. Blesa, E. Romero, and O. Nieto-Taladriz, “Artificial noise scheme to ensure secure communications in CWSN,” in Proc. 8th Int. Wireless Commun. Mobile Comput. Conf., Limassol, Cyprus, Aug. 2012, pp. 1023-1027. cited by applicant

A. Barengghi, L. Breveglieri, I. Koren, and D. Naccache, “Fault injection attacks on cryptographic devices: Theory, practice, and countermeasures”, Proc. IEEE, vol. 100, No. 11, pp. 3056-3076, Nov. 2012. cited by applicant

A. Birafane, M. El-Asmar et al., Analyzing LINC Systems, [in:] Microwave Magazine, IEEE, vol. 11, No. 5, pp. 59-71, Aug. 2010. cited by applicant

A. Gusmao, V. Goncalves, N. Esteves, “A novel approach to modeling of OQPSK type digital transmission over nonlinear radio channels”, in IEEE Journal on Selected Areas in Communications, Vo. 15, No. 4, pp. 647-655, 1997. cited by applicant

Alekajbaf Y, Masoumi N, Mohammad-Taheri M, Safavi-Naeini S. A 200 W broadband high power amplifier using hybrid power combining network and digital level control. Int J Rf Microw Comput Aided Eng. 2020. cited by applicant

Aoki, S. D. Kee, D. B. Rutledge, and A. Hajimiri, “Fully Integrated CMOS Power Amplifier Design Using the Distributed Active Transformer Architecture,” IEEE J. Solid-State Circuits, vol. 37, No. 3, pp. 371-383, Mar. 2002. cited by applicant

Astucia et al. (Efficient Amplification and Detection of Multilevel SC-FDE Signals Based on

BPSK Components, 2013 IEEE Military Communications Conference). cited by applicant

B. Schneier, "Cryptographic design vulnerabilities", IEEE Computer, vol. 31, No. 9, pp. 26-33, Sep. 1998. cited by applicant

Barton, Taylor W., et al., Theory and Implementation of RF-Input Outphasing Power Amplification, IEEE Transactions on Microwave Theory and Techniques, vol. 63, No. 12, pp. 4273-4283, Dec. 2015. cited by applicant

Benvenuto, Nevio et al., Single Carrier Modulation With Nonlinear Frequency Domain Equalization: An Idea Whose Time Has Come—Again, Proceedings of the IEEE, vol. 98, No. 1, Jan. 2010. cited by applicant

C.-F. Chou, Y.-H. Hsiao, Y.-C. Wu, Y.-H. Lin, C.-W. Wu, and H. Wang, "Design of a V-band 20-dBm wideband power amplifier using transformer-based radial power combining in 90-nm CMOS," IEEE Trans. Microwave Theory and Tech, vol. 64, No. 12, pp. 4545-4560, Dec. 2016. cited by applicant

Cao, Tao, et al., S-band 34.2dBm Envelope-Tracking Transmitter with 54% Overall Efficiency Utilizing the High Efficiency and Wideband Envelope Amplifier, IEEE 2019. cited by applicant

Cao, Wenhui, et al., A Modified Decomposed Vector Rotation-Based Behavioral Model With Efficient Hardware Implementation for Digital Predistortion of RF Power Amplifiers, IEEE Transactions on Microwave Theory and Techniques, vol. 65, No. 7, Jul. 2017. cited by applicant

Carvalho, Paulo, Marko Beko, Rui Dinis, Joao Guerreiro, and Pedro Viegas. "Apparatus for quantized linear amplification with nonlinear amplifiers." U.S. Pat. No. 10,069,467, issued Sep. 4, 2018. cited by applicant

Chen, Chi-Tsan, Kahn Envelope Elimination and Restoration Technique Using Injection-Locked Oscillators, IEEE 2012. cited by applicant

Cox, D.C., Linear Amplification with Nonlinear Components, IEEE Transactions on Communications, Dec. 1974. cited by applicant

D. Falconer, S. Ariyavisitakul, A. Benyamin-Seeyar, and B. Eidson, "Frequency domain equalization for single-carrier broadband wireless systems", Communications Magazine, IEEE, vol. 40, No. 4, pp. 58-66, Apr. 2002. cited by applicant

D. Kim, D. Kang, J. Kim, Y. Cho, and B. Kim, "Wideband envelope tracking power amplifier for LTE application", Proc. IEEE Radio Freq. Integr. Circuits Symp. (RFIC), Jun. 2012, pp. 275-278. cited by applicant

Doherty, W.H., A New High-Efficiency Power Amplifier for Modulated Waves, IEEE. cited by applicant

E. Yaacoub and M. Al-Husseini, "Achieving physical layer security with massive MIMO beamforming," 2017 11th European Conference on Antennas and Propagation (EUCAP), Paris, France, 2017, pp. 1753-1757. cited by applicant

Falconer, David, et al., Frequency Domain Equalization for Single-Carrier Broadband Wireless Systems, IEEE Communications Magazine, Apr. 2002. cited by applicant

Giannini, Franco, et al., The Doherty Amplifier: Past, Present & Future, IEEE 2015. cited by applicant

Gouba, Abel O., Y. Louet, "Theoretical analysis of the trade-off between efficiency and linearity of the high power amplifier in OFDM context", Proc. Eur. Wireless Conf., Apr. 2012, pp. 1-7. cited by applicant

Gustaffson, David, et al., A GaN MMIC Modified Doherty PA With Large Bandwidth and Reconfigurable Efficiency, IEEE Transactions on Microwave Theory and Techniques, vol. 62, No. 12, Dec. 2014. cited by applicant

H. C. Park, K. Lee and K. Feher, "Continuous phase modulation of spectrally efficient FQPSK signals", IEEE Vehicular Technology Conference, 1988, Philadelphia, PA, USA. cited by applicant

H. Tsai, W. -H. Liu, P. -C. Shen and W. -L. Huang, "A X-Band Fully-Integrated CMOS Power Amplifier using Current Combining Technique," 2019 IEEE 8th Global Conference on Consumer

Electronics (GCCE), Osaka, Japan, 2019, pp. 411-412. cited by applicant

H. Wang, C. Sideris and A. Hajimiri, "A CMOS Broadband Power Amplifier With a Transformer-Based High-Order Output Matching Network," in IEEE Journal of Solid-State Circuits, vol. 45, No. 12, pp. 2709-2722, Dec. 2010. cited by applicant

Himanshu Sharma, Neeraj Kumar, Rajkumar Tekchandani, "Physical layer security using beamforming techniques for 5G and beyond networks: A systematic review", Elsevier, Physical Communication, vol. 54, 2022. cited by applicant

Hsiao-Hwa Chen, The next generation of CDMA techniques, John Wiley & Sons, 2007. cited by applicant

J. C. Pedro, N.B.C. Carvalho, C. Fager and J. A. Garcia, "Linearity versus efficiency in mobile handset power amplifiers: A battle without a loser", Microwave Engineering Europe, pp. 19-26, Aug. 2004. cited by applicant

J. -K. Wang, Y. -H. Lin, Y. -H. Hsiao, K. -S. Yeh and H. Wang, "A V-band power amplifier with transformer combining and neutralization technique in 40-nm COMS," 2017 IEEE International Symposium on Radio-Frequency Integration Technology (RFIT), Seoul, Korea (South), 2017, pp. 113-116. cited by applicant

J. L. Massey, "An introduction to contemporary cryptology", Proc. IEEE, vol. 76, No. 5, pp. 533-549, May 1988. cited by applicant

J. S. Walling, S. -M. Yoo and D. J. Allstot, "Digital power amplifier: A new way to exploit the switched-capacitor circuit," in IEEE Communications Magazine, vol. 50, No. 4, pp. 145-151, Apr. 2012. cited by applicant

J. Wu and J. Chen, "Multiuser transmit security beamforming in wireless multiple access channels," in Proc. IEEE Int. Conf. Commun., Ottawa, Canada, ON, Jun. 2012, pp. 903-906. cited by applicant

K. Guo, P. Huang and K. Kang, "A 60-GHz 1.2 V 21 dBm power amplifier with a fully symmetrical 8-way transformer power combiner in 90 nm CMOS", IEEE MTT-S Int. Microw. Symp. Dig., Jun. 2014. cited by applicant

K. Murota and K. Hirade, "GMSK Modulation for Digital Mobile Radio Telephony", IEEE Transactions on Communications, vol. 29, Issue 7, pp. 1044-1050, Jul. 1981. cited by applicant

Kim, J., et al., Digital predistortion of wideband signals based on power amplifier model with memory, IEEE 2001. cited by applicant

Kousai, S. and A. Hajimiri, "An octave-range, watt-level, fully-integrated CMOS switching power mixer array for linearization and back-off-efficiency improvement," IEEE J. Solid-State Circuits, vol. 44, No. 12, pp. 3376-3392, Dec. 2009. cited by applicant

L. Zhang, L. -L. Kuang, Z. -Y. Ni, and J. -H. Lu, "Performance evaluation for OFDM and SC-FDE systems with high power amplifier", Proc. IET Int. Commun. Conf. Wireless Mobile Comput. (CCWMC), Dec. 2009, pp. 348-352. cited by applicant

Li, Yan, et al., Design of High Efficiency Monolithic Power Amplifier With Envelope-Tracking and Transistor Resizing for Broadband Wireless Applications, IEEE Journal of Solid-State Circuits, vol. 47, No. 9, Sep. 2012. cited by applicant

M. Bellanger, FBMC physical layer: A primer, [in] ICT-PHYDYAS, Tech. Rep., pp. 1-31, 2010. cited by applicant

M. K. Simon and D. Divsalar, "A Reduced Complexity Highly Power Bandwidth Efficient Coded Feher-Patented Quadrature-Phase-Shift-Keying System with Iterative Decoding", JPL Progress report, vol. 145, No. 42, pp. 1-17, May 2001. cited by applicant

M. K. Simon, T. Yan, "Performance Evaluation and Interpretation of Unfiltered Feher-Patented Quadrature Phase-Shift Keying (FQPSK)", California Institute of Technology, JPL-NASA Publication, TMD Progress Report California, May 15, 1999. 42-137, Pasadena. cited by applicant

M. M. Assefzadeh and A. Babakhani, "Multi-order transmission line-radial stub networks for broadband impedance matching and power combining in a watt-level silicon power amplifier,"

2016 Texas Symposium on Wireless and Microwave Circuits and Systems (WMCS), Waco, TX, USA, 2016, pp. 1-3. cited by applicant

Morgan, Dennis R., et al., A Generalized Memory Polynomial Model for Digital Predistortion of RF Power Amplifiers, IEEE Transactions on Signal Processing, vol. 54, No. 10, Oct. 2006. cited by applicant

N. Anand, S.-J. Lee, and E. W. Knightly, “Strobe: Actively securing wireless communications using zero-forcing beamforming,” in 2012 Proceedings IEEE Infocom, Mar. 2012, pp. 720-728. cited by applicant

N. Benvenuto, R. Dinis, D. Falconer, and S. Tomasin, “Single carrier modulation with nonlinear frequency domain equalization: An idea whose time has come again”, Proceedings of the IEEE, vol. 98, No. 1, pp. 69-96, Jan. 2010. cited by applicant

O. Abel Gouba, Y. Louet, “Theoretical analysis of the trade-off between efficiency and linearity of the high power amplifier in OFDM context”, Proc. Eur. Wireless Conf., Apr. 2012, pp. 1-7. cited by applicant

O. Cepheli, T. Maslak, and G. Kurt, “Analysis on the effects of artificial noise on physical layer security,” in Proc. 21st Signal Process. Commun. Appl. Conf., Haspolat, Turkey, Apr. 2013, pp. 1-4. cited by applicant

Orthogonal Waveforms and Filter Banks for Future Communication Systems, Edition: 1st, Chapter: 18, pp. 462-485, Elsevier, 2009. cited by applicant

P. Laurent, “Exact and approximate construction of digital phase modulation by superposition of amplitude modulation pulses (AMP)”, IEEE Transactions, COM-34, pp. 150-160, Feb. 1986. cited by applicant

P. Montezuma et al., “Quantized Digital Amplification with combination over the air—Achieving maximum efficiency on communication links between long range UAVs and satellites,” MILCOM 2021—2021 IEEE Military Communications Conference (MILCOM), San Diego, CA, USA, 2021, pp. 802-807. cited by applicant

P. Montezuma, A. Gusmao, “A pragmatic coded modulation choice for future broadband wireless communications”, IEEE VTC’01 (Spring), Rhodes, Greece, May 2001. cited by applicant

P. Montezuma, A. Gusmao, “Design of TC-OQAM schemes using a generalized nonlinear OQPSK-type format”, Electronic Letters, vol. 35, Issue 11, pp. 860/861, 1999. cited by applicant

P. Reynaert and A. M. Niknejad, “Power combining techniques for RF and mm-wave CMOS power amplifiers,” ESSCIRC 2007—33rd European Solid-State Circuits Conference, Munich, Germany, 2007, pp. 272-275. cited by applicant

P. Reynaert and M. Steyaert, RF Power Amplifiers for Mobile Communications, Springer, 2006. cited by applicant

P. S. K. Leung and K. Feher, “F-QPSK—a superior modulation technique for mobile and personal communications”, IEEE Trans. Broadcast., vol. 39, pp. 288-294, Jun. 1993. cited by applicant

Paulo Montezuma et al. (Power Efficient Coded 16-QAM Schemes over Nonlinear Transmitters, 34th IEEE Sarnoff Symposium, May 3-4. 2011. Princeton, NJ, USA. cited by applicant

R. Dinis, A. Gusmao and N. Esteves, “On broadband block transmission over strongly frequency-selective fading channels”, IEEE VVireless’03, Calgary, Canada, Jul. 2003. cited by applicant

R. Dinis, A. Gusmao, Nonlinear signal processing schemes for OFDM modulations within conventional or LINO transmitter structures, [in:] European Transactions on Telecommunications, Vo. 19, No., 3, pp. 257-271, Apr. 2008. cited by applicant

R. Dinis, P. Montezuma, N. Souto, J. Silva, Iterative Frequency-Domain Equalization for General Constellations, 33rd IEEE Sarnoff Symposium 2010, Princeton, USA,—Apr. 2010. cited by applicant

R. V. Nee and R. Prasad, OFDM for Wireless Multimedia Communications, 1st ed. Norwood, MA, USA: Artech House, Inc., 2000. cited by applicant

Rangan, S., T.S. Rappaport, and E. Erkip, “Millimeter-wave cellular wireless networks: Potentials

and challenges”, Proc. IEEE, vol. 102, No. 3, pp. 366-385, Mar. 2014. cited by applicant

Rappaport, T.S., et al., “Millimeter wave mobile communications for 5G cellular: it will work!”, IEEE Access, vol. 1, pp. 335-349, May 2013. cited by applicant

S. C. Cripps, RF Power Amplifiers for Wireless Communications. Artech House, 2006. cited by applicant

S. Kousai and A. Hajimiri, “An octave-range watt-level fully-integrated cmos switching power mixer array for linearization and back-off-efficiency improvement”, IEEE Journal of Solid-State Circuits, vol. 44, No. 12, pp. 3376-3392, 2009. cited by applicant

S. Shambayati, D. K. Lee, “GMSK modulation for deep space applications”, IEEE Aerospace Conference, Montana, USA, Mar. 2012. cited by applicant

Said, H.A., et al., Design and Realization of Digital Pulse Compression in Pulsed Radars Based on Linear Frequency Modulation (LFM) Waveforms Using FPGA, International Conference on Advanced Information and Communication Technology for Education (ICAICTE 2013). cited by applicant

Sim, Jaewoo, et al., Analysis and Design of Doherty Power Amplifiers for Digital Pre-distortion Linearizer, Proceedings of Asia-Pacific Microwave Conference 2007. cited by applicant

Sundeep Rangan et al. (Millimeter-Wave Cellular Wireless Networks: Potentials and Challenges, Proceedings of the IEEE I vol. 102, No. 3, Mar. 2014). cited by applicant

T. Aulin, N. Rydbeck and C. Sundberg, “Continuous phase modulation—Part II: Partial response signaling”, IEEE Transactions on Communications, Vo. COM-29, pp. 210-215, Mar. 1981. cited by applicant

T. J. Hill, “A non-proprietary, constant envelope, variant of shaped offset QPSK (SOQPSK) for improved spectral containment and detection efficiency”, IEEE Milcom Conference Record, vol. 1, pp. 347-352, California, USA, Oct. 2000. cited by applicant

T. Lee, S. Wang, C. Liu, “FQPSK-O: An improved performance constant envelope modulation scheme for OQPSK”, IEEE International Telemetering Conference, San Diego, USA, 1998. cited by applicant

T.S. Rappaport, et al., Millimeter wave mobile communications for 5g cellular: it will work, [in: IEEE Access, IEEE, vol. 1, pp. 335-249, 2013. cited by applicant

W. K. Harrison, J. Almeida, M. R. Bloch, S. W. McLaughlin, and J. Barros, “Coding for secrecy: An overview of error-control coding techniques for physical-layer security”, IEEE Signal Process. Mag., vol. 30, No. 5, pp. 41-50, Sep. 2013. cited by applicant

W. Struble, F. McGrath, K. Harrington, P. Nagle and S. Rand, “Understanding linearity in wireless communication amplifiers,” GaAs IC Symposium IEEE Gallium Arsenide Integrated Circuit Symposium. 18th Annual Technical Digest 1996, Orlando, FL, USA, 1996. cited by applicant

W. Yuan, Jeffrey S. Walling, “A Multiphase Switched Capacitor Power Amplifier” in IEEE Journal of Solid-State Circuits, vol. 52, No. 5, May 2017 pp. 1320-1330. cited by applicant

W. Yuan, V. Aparin, J. Dunworth, L. Seward and J. S. Walling, “A Quadrature Switched Capacitor Power Amplifier,” in IEEE Journal of Solid-State Circuits, vol. 51, No. 5, pp. 1200-1209, May 2016. cited by applicant

Wang, Feipeng, et al., A Monolithic High-Efficiency 2.4-GHz 20-dBm SiGe BiCMOS Envelope-Tracking OFDM Power Amplifier, IEEE Journal of Solid-State Circuits, vol. 42, No. 6, Jun. 2007. cited by applicant

Wang, Feipeng, et al., Design of Wide-Bandwidth Envelope-Tracking Power Amplifiers for OFDM Applications, IEEE Transactions on Microwave Theory and Techniques, vol. 53, No. 4, Apr. 2005. cited by applicant

Wang, Feipeng, et al., Wideband Envelope Elimination and Restoration Power Amplifier with High Efficiency Wideband Envelope Amplifier for WLAN 802.11g Applications, IEEE 2005. cited by applicant

Wang, Zhancang, Demystifying Envelope Tracking, IEEE, Mar. 6, 2015. cited by applicant

Wolf, Robert, et al., On the Maximum Efficiency of Power Amplifiers in OFDM Broadcast Systems with Envelope Following, MobiLight 2010, LNICST 45, pp. 160-170, 2010. cited by applicant

Y. Li, W. Li, Y. Wang, W. Luo, Y. Lin and H. Xu, "A Compact 144% Fractional Bandwidth CMOS Power Amplifier With an Optimization of Synthesized High-Order Matching Network," 2023 IEEE 15th International Conference on ASIC (ASICON), Nanjing, China, 2023, pp. 1-4. cited by applicant

Zhu, Anding, Decomposed Vector Rotation-Based Behavioral Modeling for Digital Predistortion of RF Power Amplifiers, IEEE Transactions on Microwave Theory and Techniques, vol. 63, No. 2, Feb. 2015. cited by applicant

Zhu, Anding, et al., Open-Loop Digital Predistorter for RF Power Amplifiers Using Dynamic Deviation Reduction-Based Volterra Series, IEEE Transactions on Microwave Theory and Techniques, vol. 56, No. 7, Jul. 2008. cited by applicant

Cappello, Tommaso, et al., Lineraziation of a 500-W L-band GaN Doherty Power Amplifier by Dual-Pulse Trap Characterization, 2019 IEEE/MTT-S International Microwave Symposium. cited by applicant

Technical White Paper, Unitary Braid Division Multiplexing High-performance, zero-attack-surface wireless, Rampart Communications. cited by applicant

Primary Examiner: Rego; Dominic E

Attorney, Agent or Firm: Kaufman & Canoles, P.C.

Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to physical layer security and more specifically to physical layer security such as that enabled by a coded decomposition of digital samples of time variant envelope signals into mapping components for energy optimization of transceiver front ends.

BACKGROUND

(2) Wireless communications are prone to interception of wirelessly transmitted signals by unauthorized wireless devices. Due to the sharing nature of a wireless communication channel, other wireless devices that are present on the same wireless communication channel at transmission may have access to the transmitted data. Accordingly, there is a need for improved techniques to avoid unauthorized or unwanted interception of transmitted data on a wireless communication channel. In addition, other desirable features and characteristics of the present disclosure will become apparent from the subsequent detailed description and embodiments, taken in conjunction with the accompanying figures and the foregoing technical field and background.

(3) The Background section of this document is provided to place embodiments of the present disclosure in technological and operational context, to assist those of skill in the art in understanding their scope and utility. Unless explicitly identified as such, no statement herein is admitted to be prior art merely by its inclusion in the Background section.

BRIEF SUMMARY OF THE DISCLOSURE

(4) The following presents a simplified summary of the disclosure in order to provide a basic understanding to those of skill in the art. This summary is not an extensive overview of the disclosure and is not intended to identify key/critical elements of embodiments of the disclosure or to delineate the scope of the disclosure. The sole purpose of this summary is to present some

concepts disclosed herein in a simplified form as a prelude to the more detailed description that is presented later.

(5) In this disclosure, a system of providing physical security with codified amplification is provided. According to one aspect, a system can receive samples $s_{\text{sub}.n} = s(t_{\text{sub}.n}) = s_{\text{sub}.I,n} + js_{\text{sub}.Q,n}$ of a baseband signal $s(t)$ at sampling instants $t_{\text{sub}.n}$ of the sampled in-phase and quadrature-phase components of the baseband signal $s(t)$, can receive an input clock signal associated with the sampling frequency $f_{\text{sub}.s} = 1/T_{\text{sub}.s}$ of the baseband signal $s(t)$, can receive a clock signal associated with an intermediate frequency $f_{\text{sub}.i}$, and can receive a clock signal associated with the carrier frequency $f_{\text{sub}.c}$ of the output signal.

(6) A quantizer circuitry can receive the samples of $s_{\text{sub}.n}$ of the baseband signal $s(t_{\text{sub}.n}) = s_{\text{sub}.I,n} + js_{\text{sub}.Q,n}$, having sampling instants $t_{\text{sub}.n}$, can compute the amplitude

$$(7) A_n = \sqrt{s_{I,n}^2 + js_{Q,n}^2}$$

and the phase

$$(8) \theta_n = \tan^{-1} \left(\frac{s_{Q,n}}{s_{I,n}} \right)$$

of each sample using a quantization encoding table with $N_{\text{sub}.b}$ bits, a table with a finite set of quantization values and a table with discrete amplitudes of quantization components $a_{\text{sub}.i} = \{a_{\text{sub}.1}, a_{\text{sub}.2}, \dots, a_{\text{sub}.N_{\text{sub}.b}}\}$, with $N_{\text{sub}.b}' = N_{\text{sub}.b}$ or $N_{\text{sub}.b}' = N_{\text{sub}.b} + 1$, in which the quantized value can be decomposed. The quantizer circuitry can generate the quantized amplitude $A_{\text{sub}.qn} = A_{\text{sub}.n} + e_{\text{sub}.q}$ of each sample, where $e_{\text{sub}.q}$ denotes the quantization error and can generate a set of quantization bits and, based on the quantization error $e_{\text{sub}.q}$, can compute a compensation factor Δ belonging to a finite discrete alphabet.

(9) A mapper circuitry can receive the amplitude $A_{\text{sub}.q}$, phase $\alpha_{\text{sub}.n}$ and compensation factor Δ for each sample, and together with the quantization table, can generate a coded mapping table with mapping codes and corresponding mapping coefficients. A secure mapper circuitry can receive the coded mapping table and the quantization table and can select the security scheme to be applied to the mapping codes and the mapping coefficients needed to generate a secure mapping table to be used in generation of bandpass components. The secure mapper circuitry can generate numerical sequences having the positions of samples in which secure mapping tables should be applied. It should be noted that the secure mapper circuitry can be configured to have numerical sequences stored in a memory element such as a look up table (LUT), read only memory (ROM), random access memory (RAM), flash memory, programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), cache memory, register memory, or the like. The secure mapper circuitry can generate the pairs of in-phase and quadrature-phase mapping components and can provide the active pair of mapping components to the corresponding bandpass modulator.

(10) The secure mapper can apply the secure mapping table with mapping coefficients $C_{\text{sub}.k'}$ where $k=1, \dots, N_{\text{sub}.b}$, with $N_{\text{sub}.b}' = N_{\text{sub}.b}$ or $N_{\text{sub}.b}' = N_{\text{sub}.b} + 1$, to create secure mapping tables with new mapping coefficients $C_{\text{sub}.k'}$ needed to achieve security on mapping the quantized value into a maximum number $N' \leq N_{\text{sub}.b}$ of active pairs of in-phase $s_{\text{sub}.I,n}$ and quadrature-phase $s_{\text{sub}.Q,n}$ components related to $N' \leq N_{\text{sub}.b}$ active mapping components given by $C_{\text{sub}.k'} \cos(\alpha_{\text{sub}.n})$ and $C_{\text{sub}.k'} \sin(\alpha_{\text{sub}.n})$.

(11) According to another aspect, the maximum number of possible bandpass components and mapping coefficients $C_{\text{sub}.k'}$ is $N_{\text{sub}.b}$, and coefficients $C_{\text{sub}.k'}$ have values from $N_{\text{sub}.b}$ discrete alphabets.

(12) According to another aspect, the maximum number of possible bandpass components or mapping coefficients $C_{\text{sub}.k'}$ is $N_{\text{sub}.b} + 1$ for $N_{\text{sub}.b}$ quantization bits and coefficients $C_{\text{sub}.k'}$ and $C_{\text{sub}.k'}$ can represent values from $N_{\text{sub}.b} + 1$ discrete alphabets. Secure mapping coefficients can be applied by bandpass modulators to generate the bandpass components required by the secure mapper circuit. The set of active bandpass components can be generated at carrier frequency $f_{\text{sub}.c}$.

- (13) According to another aspect, a set of memory elements (e.g., LUTs) can include the mapping table, the secure mapping table, the security scheme, the quantization encoding table, the sets of discrete amplitudes at, the set of quantization values, the like, or any combination thereof.
- (14) According to another aspect, a security scheme can be based on permutations of mapping codes and corresponding mapping coefficients.
- (15) According to another aspect, a security scheme can be based on phase rotations of mapping coefficients associated with mapping codes that can affect all or a subset of the mapping codes.
- (16) According to another aspect, a security scheme can be based on multiplying a scale factor and mapping coefficients associated with mapping codes that can affect all or a sub-set of the mapping codes.
- (17) According to another aspect, a security scheme can be based on puncturing mapping codes.
- (18) According to another aspect, a security scheme can be based on any combination of permutations, puncturing, phase rotations and scale factor multiplication.
- (19) According to another aspect, any of the security schemes disclosed herein can be combined or applied at the same time or applied at separate times to different samples or subsets of samples.
- (20) According to another aspect, any of the security schemes disclosed herein can be combined with interleaving techniques affecting the sequence of samples of the input signal.
- (21) According to another aspect, the secure mapping rule applied to each quantized value can be applied by a secure mapper circuitry to generate control information to enable bandpass modulators associated with the mapping components.
- (22) According to another aspect, the bandpass modulator circuitry can be configured as an in-phase/quadrature-phase (I/Q) digital-to-analog converter (I/Q DAC) or configured as an I/Q modulator.
- (23) According to another aspect, the secure mapper circuitry can provide information to control the phase of bandpass components generated by a set of bandpass modulators (I/Q modulators or I/Q DACs).
- (24) According to another aspect, the set of bandpass modulators can generate active bandpass components at the intermediate frequency $f_{sub.i}$ and a set of mixer circuitry can upconvert the active bandpass components to the carrier frequency $f_{sub.c}$.
- (25) According to another aspect, the method and apparatus for performing physical security with codified amplification includes a digital circuitry operable to compute $A_{sub.n}$ and $\alpha_{sub.n}$, quantize $A_{sub.n}$, perform the mapping encoding, define the secure mapping rule, perform the secure mapping of the quantized value into bandpass components and generate random permutation patterns, phase rotations or puncturing patterns.
- (26) According to another aspect, the method and apparatus for performing physical security with codified amplification can be implemented with analog or digital controls, including hardware, firmware or software.
- (27) According to another aspect, the method and apparatus for performing physical security with codified amplification can be implemented with analog components or with a combination of analog components and digital components.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the disclosure are shown. However, this disclosure should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Like numbers refer to like elements

throughout.

(2) FIG. 1 illustrates one embodiment of a method of providing physical layer security in accordance with various aspects as described herein.

(3) FIG. 2A illustrates one embodiment of a system of physical layer security in accordance with various aspects as described herein. FIG. 2B illustrates another embodiment of a system of physical layer security in accordance with various aspects as described herein.

(4) FIG. 3 illustrates another embodiment of a method of providing physical layer security in accordance with various aspects as described herein.

(5) FIG. 4 illustrates one embodiment of a wireless communication network in accordance with various aspects as described herein.

(6) FIG. 5 illustrates one embodiment of an electronic device in accordance with various aspects as described herein.

DETAILED DESCRIPTION

(7) For simplicity and illustrative purposes, the present disclosure is described by referring mainly to exemplary embodiments thereof. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be readily apparent to one of ordinary skill in the art that the present disclosure may be practiced without limitation to these specific details.

(8) Common security techniques rely on encryption schemes from higher layers, such as key cryptography protocols. Besides the possible vulnerability associated with the increased computational capacity of eavesdroppers, these techniques achieve security at the expense of spectral and energy inefficiencies of the transmission system. Another problem with these security techniques is the high data throughput, resulting in undesirable cryptographic or security coding with high overheads that may compromise overall efficiency and increase latency of the system.

(9) In the present disclosure, both low complexity and energy efficient PLS can be achieved, exploring the properties of coded mapping associated with signal's envelope decomposition in codified amplification. For instance, the coding rule applied to the mapping process of quantized value of signal envelope samples into bandpass components can be securely changed. Mapping codes can be varied and can be associated with mapping coefficients that when combined can generate each bandpass mapping component. Hence, mapping code changes such as permutations or puncturing of mapping codes can enable physical layer security. Further, mapping coefficients can be modified such as manipulation of the phase or amplitude of the mapping coefficients, which can further increase the degrees of freedom of possible physical layer security implementations.

(10) In the present disclosure, the quantized digital amplification in the decomposition of quantized sample values into constant envelope components can apply a coded mapping rule having different mapping codes that are associated with subsets of mapping coefficients. Hence, changes to the mapping codes and the corresponding mapping coefficients can be utilized to enable physical layer security, without additional complexity. This type of physical layer security remains undetectable as the transmission spectral mask still conforms to any standards requirements and the statistical characteristics of the signal envelope also remains Gaussian. Any transmission overhead is also avoided as well as any dependency on transmission channel statistics. Moreover, this quantized digital amplification can be applied along with a fully quantized digital amplification stage having a set of parallel amplifiers for improved efficiency or can be used with common power amplifiers employed in modern radio frequency front ends.

(11) In one exemplary embodiment, a digital signal processing implementation can generate quantized punctured bandpass components that optimize energy efficiency on power amplification and power combination. A first mapper component (e.g., software, firmware, hardware) can decompose a quantized input signal sample into a sum of bandpass components according to mapping codes and can provide a reference for a second mapper component (e.g., software, firmware, hardware) to perform a secure mapping process based on a new secure mapping rule. A

secure mapping rule can include generating permutations of the mapping codes, applying a phase rotation or a scale factor to the mapping coefficients, puncturing the mapping codes, the like, or any combination thereof. Further, a secure mapping rule can combine the several techniques and may change over the time or between subsets of the samples of a data block. The application of a secure mapping rule may not increase the complexity of a hardware-based transceiver as digital signal processing can be applied to perform the quantized digital amplification. Also, the quantized digital amplification can be applied along with a multi-amplifier power amplification stage followed by a combiner circuitry, optimized for the signal decomposition associated with the quantized digital amplification for improved energy efficiency.

(12) FIG. 1 illustrates one embodiment of a method **100** of providing physical layer security in accordance with various aspects as described herein. In FIG. 1, the method **100** can start, for instance, at block **101** where it can include receiving the samples $s_{\text{sub}.n}$ of a baseband signal $s(t)$. In one example, the samples $s_{\text{sub}.n} = s(t_{\text{sub}.n}) = s_{\text{sub}.I,n} + js_{\text{sub}.Q,n}$ of a baseband signal $s(t)$ are quantized and decomposed into several analog bandpass components using a secure mapping rule. In another example, the input signal is a baseband signal which is sampled to generate the samples to be converted posteriorly into bandpass components. In yet another example, the samples $s_{\text{sub}.I,n}$ and $s_{\text{sub}.Q,n}$ of the respective in-phase and quadrature-phase components of a baseband signal are received. In yet another example, a baseband signal is received and sampled by a sampling circuitry or by a sample and hold (S/H) circuitry to generate the samples of the input signal. In yet another embodiment, the in-phase and quadrature-phase components of a baseband signal are received. At block **102**, the method **100** can include receiving a clock reference with a sampling rate $f_{\text{sub}.s}$, receiving a clock reference with a carrier frequency $f_{\text{sub}.c}$, and receiving a clock reference with an intermediate frequency $f_{\text{sub}.i}$. At block **103**, the method **100** can include computing the amplitude $A_{\text{sub}.n}$ and the phase $\alpha_{\text{sub}.n}$ for each sample, quantizing $A_{\text{sub}.n}$ as $A_{\text{sub}.q}$ using $N_{\text{sub}.b}$ quantization bits from a quantization table, computing the quantization error $e_{\text{sub}.q}$, and computing a compensation factor Δ from $e_{\text{sub}.q}$. At block **104**, the method **100** can include providing to a mapper circuitry the values of $A_{\text{sub}.q}$, $\alpha_{\text{sub}.n}$, Δ , and the quantization encoded table. At block **105**, the method **100** can include generating a coded mapping table with mapping codes and corresponding mapping coefficients. At block **106**, the method **100** can include receiving the coded mapping table and the coded quantization table and selecting the security scheme to be applied to the mapping codes and the mapping coefficients of a secure mapping table to be applied in generation of the bandpass components. At block **107**, the method **100** can include generating a secure mapping table and numerical sequences having the positions of samples in which the secure mapping table will be applied in generation of the bandpass components. At block **108**, the method **100** can include generating the pairs of in-phase and quadrature-phase mapping components. At block **109**, the method **100** can include providing each active pair of mapping components to the corresponding bandpass modulator circuitry. At block **110**, the method **100** can include generating control information to enable or disable the bandpass modulator circuitry. At block **111**, the method **100** can include upconverting the bandpass components to an intermediate frequency $f_{\text{sub}.i}$. At block **112**, the method **100** can include upconverting the bandpass components or the intermediate frequency components $f_{\text{sub}.i}$ to a carrier frequency $f_{\text{sub}.c}$.

(13) In another embodiment, the method steps described by blocks **103** to **109** can be performed in a single method step.

(14) In another embodiment, the coded mapping table can be stored in a memory element (e.g., LUT) and as such, the method step described by block **105** is not required.

(15) In another embodiment, as understood by a skilled artisan, based on the subject matter of the current disclosure, the method steps described by blocks **101** to **109** can be performed by memory elements (e.g., LUTs) having the quantization values, the quantization bits table, the coded mapping table, the secure mapping table, the numerical sequences having the positions of samples in which the secure mapping table can be applied to generate secure pairs of mapping components

and control information.

(16) In another embodiment, the security scheme can be changed between successive samples or between successive blocks of samples.

(17) FIG. 2A illustrates one embodiment of a system **200a** of physical layer security with codified amplification in accordance with various aspects as described herein. In FIG. 2A, the system **200a** implements the method **100** of FIG. 1 with $N_{\text{sub},b}=3$ or $N_{\text{sub},b}=4$ quantization bits and the same number of mapping components. The number of mapping codes is $2^{\text{sup},N_{\text{sub},b}}$, which is associated with $N_{\text{sub},b}$ sets of mapping coefficients, each mapping coefficient with $2^{\text{sup},N_{\text{sub},b}}$ possible values. The system **200a** can receive an input signal **201-a** having the time samples $s_{\text{sub},n}=s(t_{\text{sub},n})$ of a baseband signal $s(t)$. The system **200a** can also receive a clock reference signal **201-b** having the sampling frequency $f_{\text{sub},s}$. Additionally or alternatively, the system **200a** can receive the clock reference signal **201s-o1** having an intermediate frequency $f_{\text{sub},i}$ to be applied to mixer circuitry and a reference signal clock **201-ofc** having a carrier frequency $f_{\text{sub},c}$. A skilled artisan will readily recognize that the choice of the clock reference signal can be made according to the bandwidth of the input signal, the desired frequency for the output signal, and a frequency shift of upconverting bandpass components.

(18) In another embodiment, the input signal **201-a** can include the samples $s_{\text{sub},I,n}$ and $s_{\text{sub},Q,n}$ of the respective in-phase and quadrature-phase components of a baseband signal.

(19) In another embodiment, the input signal **201-a** can be a baseband signal $s(t)$, which can be sampled by sampling circuitry **221**, to generate the time samples $s_{\text{sub},n}=s(t_{\text{sub},n})$.

(20) The scope of this disclosure can include the application of other clock references or other input signals, and the implementation of such variations will be readily apparent to a skilled artisan. The clock signal **201-b** can be a clock reference signal having the sampling frequency $f_{\text{sub},s}$ and can be input to the sampling circuitry **221**, the quantizer circuitry **203**, the mapper circuitry **208**, the secure mapper circuitry **211**, the set of bandpass modulator circuitry **213**- $\{1, \dots, N_{\text{sub},b}+1\}$, the combiner circuitry **217**, or the like.

(21) The quantizer circuitry **203**, for each sample $s_{\text{sub},n}$, can determine the amplitude

$$(22) A_n = \sqrt{s_{I,n}^2 + s_{Q,n}^2}$$

and the phase

$$(23) \theta_n = \tan^{-1} \left(\frac{s_{Q,n}}{s_{I,n}} \right).$$

The quantizer circuitry **203** can apply a quantization encoding table having $N_{\text{sub},b}$ quantization bits, can apply a table having the quantization values, and a table having the set of discrete amplitudes of quantization components $a_{\text{sub},i}=\{a_{\text{sub},1}, a_{\text{sub},2}, \dots, a_{\text{sub},N_{\text{sub},b}}\}$ in which any quantized value can be decomposed. For each sample $s_{\text{sub},n}$, the quantizer circuitry **203** can quantize the amplitude $A_{\text{sub},n}$, can generate the quantized amplitude $A_{\text{sub},qn}=A_{\text{sub},n}+e_{\text{sub},q}$ and the set of quantization amplitude bits **207**- $\{1, \dots, N_{\text{sub},b}\}$, and can determine a quantization error compensation factor Δ belonging to a finite discrete alphabet based on the quantization error $e_{\text{sub},q}$. Further, the quantizer circuitry **203** can generate and send the quantization amplitude table **204** to the mapper circuitry **208**. In addition, the quantizer circuitry **203** can send, to the mapper circuitry **208**, the quantized amplitude bits **207**- $\{1, \dots, N_{\text{sub},b}\}$, the quantized amplitude $A_{\text{sub},q}$ **205-1**, the phase $\alpha_{\text{sub},n}$ **205-2** and the amplitude quantization error Δ **205-3**, and the discrete amplitudes of quantization components $a_{\text{sub},i}$ **206**.

(24) In FIG. 2A, the mapper circuitry **208** can determine a coded mapping table having mapping codes and corresponding mapping coefficients. The mapper circuitry **208** can apply the quantization table to define the set of mapping codes $M_{\text{sub},i}=\{M_{\text{sub},0}; \dots; M_{\text{sub},N_{\text{sub},c}}\}$, with $N_{\text{sub},c}=2^{\text{sup},N_{\text{sub},b}}-1$, which can be associated with the sets of mapping coefficients $C_{\text{sub},N_{\text{sub},b},\text{sup},(0)} \dots C_{\text{sub},1,\text{sup},(0)}, \dots, C_{\text{sub},N_{\text{sub},b},\text{sup},(N_{\text{sub},c},\text{sup},)} \dots C_{\text{sub},1,\text{sup},(N_{\text{sub},c},\text{sup},)}$ and can be defined in terms of the mapping coefficients $C_{\text{sub},k}=\{C_{\text{sub},k,\text{sup},(1)}; \dots; C_{\text{sub},k,\text{sup},(N_{\text{sub},c},\text{sup},)}\}$, with $k=1, \dots, N_{\text{sub},b}$. For example, with $N_{\text{sub},b}=3$ quantization

bits, the mapper circuitry **208** can define an encoded mapping as shown in Table I below, with eight (8) mapping codes $M_{sub.i} = \{M_{sub.0}; \dots; M_{sub.7}\}$ directly related to the sets of mapping coefficients $C_{sub.3.sup.}(0)C_{sub.2.sup.}(0)C_{sub.1.sup.}(0)$, $C_{sub.3.sup.}(1)C_{sub.2.sup.}(1)C_{sub.1.sup.}(1)$, $\dots$, $C_{sub.3.sup.}(7)C_{sub.2.sup.}(7)C_{sub.1.sup.}(7)$.

(25) TABLE-US-00001 TABLE I Mapping Coefficients and Mapping Codes for $N_{sub.b} = 3$

$C_{sub.3}$	$C_{sub.2}$	$C_{sub.0}$	Mapping Code $M_{sub.i}$	$C_{sub.3.sup.}(0)$	$C_{sub.2.sup.}(0)$	$C_{sub.1.sup.}(0)$
$M_{sub.0}$	$C_{sub.3.sup.}(1)$	$C_{sub.2.sup.}(1)$	$C_{sub.1.sup.}(1)$	$M_{sub.1}$	$C_{sub.3.sup.}(2)$	$C_{sub.2.sup.}(2)$
$C_{sub.1.sup.}(2)$	$M_{sub.2}$	$C_{sub.3.sup.}(3)$	$C_{sub.2.sup.}(3)$	$C_{sub.1.sup.}(3)$	$M_{sub.3}$	$C_{sub.3.sup.}(4)$
$C_{sub.2.sup.}(4)$	$C_{sub.1.sup.}(4)$	$M_{sub.4}$	$C_{sub.3.sup.}(5)$	$C_{sub.2.sup.}(5)$	$C_{sub.1.sup.}(5)$	$M_{sub.5}$
$C_{sub.3.sup.}(6)$	$C_{sub.2.sup.}(6)$	$C_{sub.1.sup.}(6)$	$M_{sub.6}$	$C_{sub.3.sup.}(7)$	$C_{sub.2.sup.}(7)$	$C_{sub.1.sup.}(7)$
$M_{sub.7}$						

(26) $C_{sub.1}$ mapping coefficients can represent values from a set of amplitudes $C_{sub.1.sup.}(i)$, $i=0, \dots, 7$ that belongs to a discrete alphabet $A_{sub.1}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{sub.1}$ or the amplitude quantization error Δ . $C_{sub.2}$ mapping coefficients can represent values from a set of amplitudes $C_{sub.2}(i)$, $i=0, \dots, 7$ that belongs to a discrete alphabet $A_{sub.2}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{sub.1}$, the discrete amplitudes of the second quantization component $a_{sub.2}$ or the amplitude quantization error Δ . $C_{sub.3}$ mapping coefficients can represent values from a set of amplitudes $C_{sub.3.sup.}(i)$, $i=0, \dots, 7$ that belongs to a discrete alphabet $A_{sub.3}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{sub.1}$, the discrete amplitudes of the third quantization component $a_{sub.3}$, or the amplitude quantization error Δ .

(27) In another exemplary embodiment with $N_{sub.b}=4$ quantization bits and $N_{sub.b} = N_{sub.b}$ possible bandpass components the mapper circuitry **208** can apply the quantization bits and the quantization table **206** to define an encoded mapping table with mapping codes $M_{sub.i} = \{M_{sub.0}; \dots; M_{sub.15}\}$ directly related to the sets of mapping coefficients $C_{sub.4.sup.}(0)C_{sub.3.sup.}(0)C_{sub.2.sup.}(0)C_{sub.1.sup.}(0)$, $C_{sub.4.sup.}(1)C_{sub.3.sup.}(1)C_{sub.2.sup.}(1)C_{sub.1.sup.}(1)$, $\dots$, $C_{sub.4.sup.}(15)C_{sub.3.sup.}(15)C_{sub.2.sup.}(15)C_{sub.1.sup.}(15)$. $C_{sub.1}$ mapping coefficients can represent values from a set of amplitudes $C_{sub.1.sup.}(i)$, $i=0, \dots, 15$ that belong to a discrete alphabet $A_{sub.1}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{sub.1}$, or the amplitude quantization error Δ . $C_{sub.2}$ mapping coefficients can represent values from a set of amplitudes $C_{sub.2.sup.}(i)U$, $i=0, \dots, 15$ that belong to a discrete alphabet $A_{sub.2}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{sub.1}$, the discrete amplitudes of the second quantization component $a_{sub.2}$, or the amplitude quantization error Δ . $C_{sub.3}$ mapping coefficients can represent values from a set of amplitudes $C_{sub.3.sup.}(i)$, $i=0, \dots, 15$ that belong to a discrete alphabet $A_{sub.3}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{sub.1}$, the discrete amplitudes of the second quantization component $a_{sub.2}$, the discrete amplitudes of the third quantization component $a_{sub.3}$, or the amplitude quantization error Δ . $C_{sub.4}$ mapping coefficients can represent values from a set of amplitudes $C_{sub.4.sup.}(i)$, $i=0, \dots, 15$ that belong to a discrete alphabet $A_{sub.4}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{sub.1}$, the discrete amplitudes of the second quantization component $a_{sub.2}$, the discrete amplitudes of the fourth quantization component $a_{sub.4}$, or the amplitude quantization error Δ .

(28) From the mapper circuitry **208**, the secure mapper circuitry **211** can receive the mapping codes **209**- $\{1, \dots, N_{sub.c}\}$ and the mapping coefficients **210**- $\{1, \dots, N_{sub.b}\}$ and can select the security scheme to apply in determining the updated secure mapping codes and the corresponding mapping coefficients. The secure mapper circuitry **211** can apply the security scheme to determine the secure mapping coefficients $C_{sub.k'}$ where $k=1, \dots, N_{sub.b'}$, and can generate secure

mapping tables with new mapping coefficients $C_{k'}$ required to achieve security in mapping the quantized amplitude value into a maximum number $N' \leq N_{b,n}$, of in-phase $s_{I,n}$ and quadrature-phase $s_{Q,n}$ mapping components among the $N_{b,n}$ possible values $2^{12-\{1_{i,q}, \dots, N_{b,q}\}}$, $2^{12-\{1_{i,q}, \dots, N_{b,q}\}}$, related to $N' \leq N_0$, active mapping components given by $C_{k'} \cos(\alpha_n)$ and $C_{k'} \sin(\alpha_n)$, respectively.

(29) In another embodiment, a security scheme can include permutations between mapping codes, phase rotations of mapping coefficients, puncturing of mapping codes and coefficients, scale factor multiplication of mapping coefficients, the like, or any combination thereof. A secure mapper circuitry can generate a secure mapping table with secure mapping codes and corresponding mapping coefficients to be applied to the mapping of quantized values and can generate the number sequence with the samples in which the secure mapping rule should be applied.

(30) In another embodiment, with permutations of mapping codes, a permutation array $P = \{P_0, P_1, \dots, P_{N_c}\}$, being the values adopted to define a set of permuted mapping codes $M_{i'} = \{M_{P_0}; M_{P_1}; \dots; M_{P_{N_c}}\}$, in which mapping codes are associated with the corresponding sets of secure mapping coefficients $C_{N_{b,n}^{sup}}$. ($P_{i^{sup}}$), $C_{N_{b,n}^{sup}-1^{sup}(P_{i^{sup}})}$, $\dots$, $C_{1^{sup}(P_{i^{sup}})}$, with $i=0, \dots, N_{b,n}$.

(31) In one exemplary embodiment, with $N_{b,n}=3$ and $N_{c,n}=8$ mapping codes, and for a permutation array of $P = \{P_0, P_1, \dots, P_7\}$, results in a new permuted set of mapping codes $M_{i'} = \{M_{P_0}; M_{P_1}; \dots; M_{P_7}\}$, which are associated with the corresponding sets of mapping coefficients given by $C_{3^{sup}(P_0^{sup})}C_{2^{sup}(P_0^{sup})}C_{1^{sup}(P_0^{sup})}$, $C_{3^{sup}(P_1^{sup})}C_{2^{sup}(P_1^{sup})}C_{1^{sup}(P_1^{sup})}$, $\dots$, $C_{3^{sup}(P_7^{sup})}C_{2^{sup}(P_7^{sup})}C_{1^{sup}(P_7^{sup})}$. For instance, a permutation array of $P = \{7, 0, 6, 1, 4, 2, 5, 3\}$ results in Table II below.

(32) TABLE-US-00002 TABLE II Example of Mapping Table After Permutation of Mapping Codes

Codes	$C_{3^{sup}}$	$C_{2^{sup}}$	$C_{1^{sup}}$	Mapping Code	$M_{i'}$	$C_{3^{sup}(p_0)}$	$C_{2^{sup}(p_0)}$
$C_{1^{sup}(p_0)}$	$M'_{sub.0}$	$M_{sub.0}$	$M_{sub.7}$	$C_{3^{sup}(p_1)}$	$C_{2^{sup}(p_1)}$		
$C_{1^{sup}(p_1)}$	$M'_{sub.1}$	$M_{sub.1}$	$M_{sub.0}$	$C_{3^{sup}(p_2)}$	$C_{2^{sup}(p_2)}$		
$C_{1^{sup}(p_2)}$	$M'_{sub.2}$	$M_{sub.2}$	$M_{sub.6}$	$C_{3^{sup}(p_3)}$	$C_{2^{sup}(p_3)}$		
$C_{1^{sup}(p_3)}$	$M'_{sub.3}$	$M_{sub.3}$	$M_{sub.1}$	$C_{3^{sup}(p_4)}$	$C_{2^{sup}(p_4)}$		
$C_{1^{sup}(p_4)}$	$M'_{sub.4}$	$M_{sub.4}$	$M_{sub.4}$	$C_{3^{sup}(p_5)}$	$C_{2^{sup}(p_5)}$		
$C_{1^{sup}(p_5)}$	$M'_{sub.5}$	$M_{sub.5}$	$M_{sub.2}$	$C_{3^{sup}(p_6)}$	$C_{2^{sup}(p_6)}$		
$C_{1^{sup}(p_6)}$	$M'_{sub.6}$	$M_{sub.6}$	$M_{sub.5}$	$C_{3^{sup}(p_7)}$	$C_{2^{sup}(p_7)}$		
$C_{1^{sup}(p_7)}$	$M'_{sub.7}$	$M_{sub.7}$	$M_{sub.3}$				

(33) With eight (8) mapping codes, the number of possible permutations is forty thousand, three hundred and twenty (40,320) and permutations can change dynamically over each data block or within a data block according to a predefined pattern.

(34) In another exemplary embodiment, with $N_{b,n}=4$ and $N_{c,n}=16$ mapping codes, for a permutation array $P = \{P_0, P_1, \dots, P_{15}\}$, results in a new permuted set of mapping codes $M_{i'} = \{M_{P_0}; M_{P_1}; \dots; M_{P_{15}}\}$, which are associated with the corresponding sets of mapping coefficients given by $C_{4^{sup}(0)}C_{3^{sup}(0)}C_{2^{sup}(0)}C_{1^{sup}(0)}$, $C_{4^{sup}(1)}C_{3^{sup}(1)}C_{2^{sup}(1)}C_{1^{sup}(1)}$, $\dots$, $C_{4^{sup}(15)}C_{3^{sup}(15)}C_{2^{sup}(15)}C_{1^{sup}(15)}$.

(35) In another embodiment, the secure mapper circuitry can apply a puncturing array $P_u = \{p_{u,1}, \dots, p_{u,N_{c,n}}\}$, where each $p_{u,t}$ represents a value of one (1) without puncturing or a value of X with puncturing, with the puncturing pattern to be applied to puncture mapping codes $M_{i'} = \{M_{sub.0}; M_{sub.1}; \dots; M_{sub.N_{c,n}}\}$. For instance, with $N_{b,n}=3$ and a puncturing rule of $P_u = \{1, X, 1, 1, X, 1, 1\}$, results in the set of punctured mapping codes $M_{i'} = \{M_{sub.0}; M_{sub.1}; \dots; M_{sub.7}\}$, directly related to the sets of mapping coefficients $C_{3^{sup}}$.

(0)C.sub.2.sup.(0)C.sub.1.sup.(0), X, C.sub.3.sup.(2)C.sub.2.sup.(2)C.sub.1.sup.(2), X, C.sub.3.sup.(3)C.sub.2.sup.(3)C.sub.1.sup.(3), X, C.sub.3.sup.(5)C.sub.2.sup.(5)C.sub.1.sup.(5), C.sub.3.sup.(6)C.sub.2.sup.(6)C.sub.1.sup.(6), C.sub.3.sup.(7)C.sub.2.sup.(7)C.sub.1.sup.(7).

(36) In another exemplary embodiment, with $N_{sub.b}=4$ quantization bits and $N_{sub.b} = N_{sub.b}$ possible bandpass components for a puncturing rule $P_u=\{1, X, 1, 1, X, 1, 1, 1, 1, 1, X, 1, 1, 1, X, 1\}$, results in the set of mapping codes $M_{sub.i}'=\{M_{sub.0}; X; M_{sub.2}; M_{sub.3}; X; M_{sub.5}; \dots; M_{sub.9}; X; M_{sub.11}; M_{sub.12}; M_{sub.13}; X; M_{sub.15}\}$, directly related to the sets of mapping coefficients: C.sub.4.sup.(0)C.sub.3.sup.(0)C.sub.2.sup.(0)C.sub.1.sup.(0), X, C.sub.4.sup.(2)C.sub.3.sup.(2)C.sub.2.sup.(2)C.sub.1.sup.(2), X, C.sub.4.sup.(3)C.sub.3.sup.(3)C.sub.2.sup.(3)C.sub.1.sup.(3), X, C.sub.4.sup.(5)C.sub.3.sup.(5)C.sub.2.sup.(5)C.sub.1.sup.(5), $\dots$, C.sub.4.sup.(9)C.sub.3.sup.(9)C.sub.2.sup.(9)C.sub.1.sup.(9), X, C.sub.4.sup.(11)C.sub.3.sup.(11)C.sub.2.sup.(11)C.sub.1.sup.(11), $\dots$, C.sub.4.sup.(13)C.sub.3.sup.(13)C.sub.2.sup.(13)C.sub.1.sup.(13), X, C.sub.4.sup.(15)C.sub.3.sup.(15)C.sub.2.sup.(15)C.sub.1.sup.(15).

(37) In another embodiment, the puncturing rule can be constant over the entire data block or may differ for different subsets of samples within the data block.

(38) In another embodiment, mapping codes can be punctured according to a predetermined rule.

(39) In another embodiment, the puncturing rule can be applied as a random phase rotation $C_{sub.k}.sup.(i)=e^{sup.j}\alpha C_{sub.k}.sup.(i)$ to a sub-set of mapping codes and corresponding mapping coefficients, with α denoting the phase rotation value. A phase rotation can affect all mapping codes or a subset of mapping codes. A phase rotation value can be predetermined, or the phase rotation may change over the data block or samples of the envelope according to a predetermined rule. This corresponds to a phase rotation of the values of the alphabets $A_{sub.k}$ associated with the sets of possible values of mapping coefficients.

(40) In another embodiment, a phase rotation can be applied to only a set of mapping coefficients $C_{sub.k}.sup.(i)=e^{sup.j}\alpha C_{sub.k}.sup.(i)$, for $i=i_{sub.1}, \dots, i_{sub.n}$ and $k=1, \dots, N'$, with $N'<N_{sub.b}$, with only some alphabets of possible values of mapping coefficients being affected by phase rotation.

(41) In another embodiment, a scale factor can be applied to a limited set of mapping coefficients $C_{sub.ks}.sup.(i)=s_{sub.f}C_{sub.ks}.sup.(i)$, for $i=i_{sub.1}, \dots, i_{sub.n}$ and $k_{sub.s}=k_{sub.1}, k_{sub.2}, \dots, N'$, with $N'<N_{sub.b}$.

(42) In another embodiment, for a security scheme based on permutations of mapping codes, any combination of puncturing, phase rotations and multiplication of mapping coefficients by a scale factor can be applied to different subsets of samples.

(43) In FIG. 2A, the secure mapper **211** can generate a set of $N'\leq N_{sub.b}$ active pairs of in-phase and quadrature-phase components related to $N'\leq N_{sub.b}$ mapping components, based on a secure mapping table such as stored in a memory element (e.g., LUT) that contains the mapping coefficient values of the secure mapping rule.

(44) In another embodiment, the secure mapping table can also include the quantization bits, the sets of discrete amplitudes $a_{sub.i}$ and the set of quantized values.

(45) In another embodiment, the mapping table, the secure mapping table, the quantization encoding table, the quantization values table and quantization amplitude components can be previously stored in memory elements (e.g., LUTs) that are available to the quantizer circuitry, the mapper, or the secure mapper circuit.

(46) In another embodiment, memory elements (e.g., LUTs) can include the permutation rules, the puncturing rules, the phase rotations rules to be applied in the definition of the secure mapping tables that are also previously stored in those memory elements.

(47) In another embodiment, the secure mapper circuitry **211** can generate numerical sequences having the positions of samples in which secure mapping tables are applied, or numerical sequences are previously stored in memory elements (e.g., LUTs).

(48) In another embodiment, sets of numerical sequences with the positions of samples affected by

security schemes can be previously stored in LUTs.

(49) In FIG. 2A, the secure mapper **211** can output $N' \leq N_{\text{sub.b}}$ active pairs of in-phase and quadrature-phase mapping components **212**-{1.sub.i, . . . , N.sub.bi}, **212**-{1.sub.q, . . . , N.sub.bq}, to a set of N' bandpass modulator circuitry **213**-{1, . . . , N.sub.b}. The set of N' bandpass modulator circuitry **213**-{1, . . . , N.sub.b} can be configured as I/Q DAC circuitry or I/Q modulator circuitry, can generate N' active bandpass components **214**-{1.sub.i, . . . , N'}. The set of N' bandpass modulator circuitry **213**-{1, . . . , N.sub.b} can generate the set of bandpass components **214**-{1.sub.i, . . . , N'} having the carrier frequency $f_{\text{sub.c}}$. The secure mapper circuitry **211** can also generate control information to activate or deactivate the mixer circuitry **215**-{O.sub.1, . . . , O.sub.N.sub.b}.

(50) In another embodiment, the secure mapper circuitry **211** can provide control information to the set of active bandpass modulator circuitry **213**-{1, . . . , N.sub.b} to control the initial phase of the generated bandpass components **214**-{1.sub.i, . . . , N'}.

(51) In another embodiment, the set of bandpass modulator circuitry **213**-{1, . . . , N.sub.b} can generate a set of N' active bandpass components **214**-{1.sub.i, . . . , N'} having the intermediate frequency $f_{\text{sub.i}}$ and the mixer circuitry **215**-{O.sub.1, . . . , O.sub.N.sub.b} can upconvert the bandpass components **214**-{1.sub.i, . . . , N'} to the carrier frequency $f_{\text{sub.c}}$ to obtain the carrier frequency components **216**-{1, . . . , N.sub.b}.

(52) In FIG. 2A, the set of mixer circuitry **215**-{O.sub.1, . . . , O.sub.N.sub.b} can receive signal **201s-o1** having a certain reference frequency.

(53) In another embodiment, the set of mixer circuitry **215**-{O.sub.1, . . . , O.sub.N.sub.b} can receive signal **201-ofc** having the carrier frequency $f_{\text{sub.c}}$.

(54) In another embodiment, the certain reference signal is associated with the intermediate frequency $f_{\text{sub.i}}$.

(55) In another embodiment, the certain reference signal is associated with up conversion from the intermediate frequency $f_{\text{sub.i}}$ to the carrier frequency $f_{\text{sub.c}}$.

(56) In another embodiment, the bandpass modulator circuitry **213**-{1, . . . , N.sub.b} can output feedback signals **F**-{O.sub.1, . . . , O.sub.N.sub.b} to the secure mapper circuitry **211** to generate time or phase control information to enable time or phase synchronization of the active bandpass components **214**-{1.sub.i, . . . , N'}.

(57) In another embodiment, the mixer circuitry **215**-{O.sub.1, . . . , O.sub.N.sub.b} can output the feedback signals **F**-{O.sub.1, . . . , O.sub.N.sub.b} to the secure mapper circuitry **211** to generate time or phase control information to enable time or phase synchronization of the active bandpass components **214**-{1.sub.i, . . . , N'}.

(58) In another embodiment, an output circuitry **225** can include a combiner circuitry **217**. The combiner circuitry **217** can combine the components **216**-{1, . . . , N.sub.b} to obtain the output signal **218**.

(59) In another exemplary embodiment, the output circuitry **225** can include a set of amplifiers (e.g., power amplifiers) with each amplifier corresponding to a certain component signal **216**-{1, . . . , N.sub.b}. The set of amplifiers can amplify the components **216**-{1, . . . , N.sub.b} to obtain a set of amplified signals. The combiner circuitry **225** can then combine the set of amplified signals to obtain the output signal **218**.

(60) In another embodiment, the sampling circuitry **221**, the mapper circuitry **208**, the secure mapper circuitry **211**, and the quantizer circuitry **203** can be configured in a single circuitry having hardware, software or firmware, including digital or analog circuitry with or without a microprocessor, a field programmable gate array (FPGA), or a digital signal processor (DSP).

(61) In another embodiment, the mapper circuitry **208** and the secure mapper circuitry **211** can be configured in a single circuitry **223** having hardware, software or firmware, including digital or analog circuitry with or without a microprocessor, an FPGA, a DSP, or the like.

(62) FIG. 2B illustrates one embodiment of a system **200b** of physical layer security in accordance

with various aspects as described herein. In FIG. 2B, the system **200b** implements the method **100** of FIG. 1 with $N_{\text{sub}.b}$ quantization bits and $N_{\text{sub}.b'} = N_{\text{sub}.b} + 1$ possible bandpass components $214 - \{1_{\text{sub}.i}, \dots, N_{\text{sub}.b+1}\}$. The clock signal **201-b** can be a clock reference signal having the sampling frequency $f_{\text{sub}.s}$ and can be input to the sampling circuitry **221**, the quantizer circuitry **203**, the mapper circuitry **208**, the secure mapper circuitry **211**, the set of bandpass modulator circuitry **213** - $\{1, \dots, N_{\text{sub}.b+1}\}$, the combiner circuitry **217**, the like, or any combination thereof.

(63) In FIG. 2B, the mapper circuitry **208** generates a set of $N' \leq N_{\text{sub}.b'}$ active pairs of in-phase and quadrature-phase components related to $N' \leq N_{\text{sub}.b'}$ mapping components according to a mapping table, stored in a memory element (e.g., LUT) configured to include the mapping codes and mapping coefficients associated with the bandpass components, and may also include the quantization bit combinations, the sets of discrete amplitudes $a_{\text{sub}.i}$, and the set of quantized values. In this exemplary embodiment, the system **200b** is configured for $N_{\text{sub}.b} = 4$ quantization bits and $N_{\text{sub}.b'} = N_{\text{sub}.b} + 1$ possible bandpass components. Other embodiments can be implemented employing an alternative number of quantization bits. The mapper circuitry **208** can include quantization bits and a quantization encoding table to define a mapping table with mapping codes. The mapper circuitry **208** can include the quantization table to define the set of mapping codes $M_{\text{sub}.i} = \{M_{\text{sub}.0}; M_{\text{sub}.1}; \dots; M_{\text{sub}.N_{\text{sub}.c}}\}$, with $N_{\text{sub}.c} = 2^{\text{sup}.N_{\text{sub}.b}} - 1$, which are associated with the sets of mapping coefficients $C_{\text{sub}.N_{\text{sub}.b'}. \text{sup}.(0)} \dots C_{\text{sub}.1. \text{sup}.(0)}, \dots, C_{\text{sub}.N_{\text{sub}.b'}. \text{sup}.(N_{\text{sub}.c. \text{sup}.})} \dots C_{\text{sub}.1. \text{sup}.(N_{\text{sub}.c. \text{sup}.})}$ and defined in terms of mapping coefficients $C_{\text{sub}.k} = \{C_{\text{sub}.k. \text{sup}.(1)}; \dots; C_{\text{sub}.k. \text{sup}.(N_{\text{sub}.c. \text{sup}.})}\}$, with $k = 1, \dots, N_{\text{sub}.b'}$. For example, with $N_{\text{sub}.b} = 4$ quantization bits and $N_{\text{sub}.b'} = 5$ results in $N_{\text{sub}.c} = 16$ mapping codes $M_{\text{sub}.i} = \{M_{\text{sub}.0}; \dots; M_{\text{sub}.15}\}$ directly related to the sets of mapping coefficients $C_{\text{sub}.5. \text{sup}.(0)} C_{\text{sub}.4. \text{sup}.(0)} C_{\text{sub}.3. \text{sup}.(0)} C_{\text{sub}.2. \text{sup}.(0)} C_{\text{sub}.1. \text{sup}.(0)}, C_{\text{sub}.5. \text{sup}.(1)} C_{\text{sub}.4. \text{sup}.(1)} C_{\text{sub}.3. \text{sup}.(1)} C_{\text{sub}.2. \text{sup}.(1)} C_{\text{sub}.1. \text{sup}.(1)}, \dots, C_{\text{sub}.5. \text{sup}.(15)} C_{\text{sub}.4. \text{sup}.(15)} C_{\text{sub}.3. \text{sup}.(15)} C_{\text{sub}.2. \text{sup}.(15)} C_{\text{sub}.1. \text{sup}.(15)}$. The coefficients $C_{\text{sub}.k}$, $k = 1, \dots, N_{\text{sub}.b} + 1$, when security is not applied, can map the quantized magnitude value into $N' \leq N_{\text{sub}.b}$ pairs of in-phase $s_{\text{sub}.I,n}$ and quadrature-phase $s_{\text{sub}.Q,n}$ mapping components. $C_{\text{sub}.1}$ mapping coefficients can represent values from a set of amplitudes $C_{\text{sub}.1. \text{sup}.(i)}$, $i = 0, \dots, 15$, belonging to a discrete alphabet $A_{\text{sub}.1}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{\text{sub}.1}$ or the amplitude quantization error Δ . $C_{\text{sub}.2}$ mapping coefficients can represent values from a set of amplitudes $C_{\text{sub}.2. \text{sup}.(i)}$, $i = 0, \dots, 15$, belonging to a discrete alphabet $A_{\text{sub}.2}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{\text{sub}.1}$, or the amplitude quantization error Δ . $C_{\text{sub}.3}$ mapping coefficients can represent values from a set of amplitudes $C_{\text{sub}.3. \text{sup}.(i)}$, $i = 0, \dots, 15$, belonging to a discrete alphabet $A_{\text{sub}.3}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{\text{sub}.1}$, the discrete amplitudes of the second quantization component $a_{\text{sub}.2}$, or the amplitude quantization error Δ . $C_{\text{sub}.4}$ mapping coefficients can represent values from a set of amplitudes $C_{\text{sub}.4. \text{sup}.(i)}$, $i = 0, \dots, 15$, belonging to a discrete alphabet $A_{\text{sub}.4}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{\text{sub}.1}$, the discrete amplitudes of the second quantization component $a_{\text{sub}.2}$, the discrete amplitudes of the third quantization component $a_{\text{sub}.3}$, or the amplitude quantization error Δ . $C_{\text{sub}.5}$ mapping coefficients can represent values from a set of amplitudes $C_{\text{sub}.5. \text{sup}.(i)}$, $i = 0, \dots, 15$, belonging to a discrete alphabet $A_{\text{sub}.5}$ with a null value and other values defined as a function of the discrete amplitudes of the first quantization component $a_{\text{sub}.1}$, the discrete amplitudes of the second quantization component $a_{\text{sub}.2}$, the discrete amplitudes of the third quantization component $a_{\text{sub}.3}$, the discrete amplitudes of the fourth quantization component $a_{\text{sub}.4}$, or the amplitude quantization error Δ .

(64) In another embodiment, when security is not applied, the coefficients $C_{\text{sub}.k}$, $k = 1, \dots, N_{\text{sub}.b} + 1$ can map the quantized magnitude value into $N' \leq N_{\text{sub}.b}$ pairs of in-phase $s_{\text{sub}.I,n}$ and

quadrature-phase s.sub.Q,n mapping components. C.sub.1 mapping coefficients can represent values from a set of amplitudes C.sub.1.sup.(i), i=0, . . . , 15, belonging to a discrete alphabet A.sub.1 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a.sub.1 or the amplitude quantization error Δ . C.sub.2 mapping coefficients can represent values from a set of amplitudes C.sub.2.sup.(i), i=0, . . . , 15, belonging to a discrete alphabet A.sub.2 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a.sub.1, the discrete amplitudes of the second quantization component a.sub.2, or the amplitude quantization error Δ . C.sub.3 mapping coefficients can represent values from a set of amplitudes C.sub.3.sup.(i), i=0, . . . , 15, belonging to a discrete alphabet A.sub.3 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a.sub.1, the discrete amplitudes of the second quantization component a.sub.2, the discrete amplitudes of the third quantization component a.sub.3, or the amplitude quantization error Δ . C.sub.4 mapping coefficients can represent values from a set of amplitudes C.sub.4.sup.(i), i=0, . . . , 15, belonging to a discrete alphabet A.sub.4 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a.sub.1, the discrete amplitudes of the second quantization component a.sub.2, the discrete amplitudes of the third quantization component a.sub.3, the discrete amplitudes of the fourth quantization component a.sub.4, or the amplitude quantization error Δ . C.sub.5 mapping coefficients can represent values from a set of amplitudes C.sub.5.sup.(i), i=0, . . . , 15, belonging to a discrete alphabet A.sub.5 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a.sub.1, the discrete amplitudes of the second quantization component a.sub.2, the discrete amplitudes of the third quantization component a.sub.3, the discrete amplitudes of the fourth quantization component a.sub.4, the discrete amplitudes of the fifth quantization component a.sub.5, or the amplitude quantization error Δ .

(65) In FIG. 2B, the secure mapper circuitry **211** can receive from the mapper circuitry **208** the mapping codes **209**-{1, . . . , N.sub.c} and the mapping coefficients **210**-{1, . . . , N.sub.b+1} and can select the security scheme to apply to the new secure mapping codes and corresponding mapping coefficients. The secure mapper circuitry **211** can apply the security scheme to compute the secure mapping coefficients C.sub.k', where k=1, . . . , N.sub.b', where N.sub.b'=N.sub.b+1, and can generate the secure mapping tables with new mapping coefficients C.sub.k' needed to achieve security on mapping the quantized value into a maximum number $N' \leq N.sub.b'$ among the N.sub.b' possible values of in-phase s.sub.I,n and quadrature-phase s.sub.Q,n mapping components **212**-{1.sub.i, . . . , N.sub.b'i}, **212**-{1.sub.q, . . . , N.sub.b'i}, related to $N' \leq N_E$ active mapping components given by C.sub.k' cos($\alpha.sub.n$) and C.sub.k' sin($\alpha.sub.n$).

(66) In one embodiment, a security scheme can include permutations between mapping codes, phase rotations of mapping coefficients, puncturing of the mapping codes and the mapping coefficients, or scale factor multiplication of the mapping coefficients. The secure mapper circuitry can generate a secure mapping table having the secure mapping codes and corresponding mapping coefficients configured to map the quantized values into bandpass components and can generate numerical sequences with the positions of samples in which secure mapping tables are applied.

(67) In another embodiment, with permutations of mapping codes, a permutation array {P.sub.0, P.sub.1, . . . , P.sub.N.sub.c}, being the values adopted to define a set of permuted mapping codes M.sub.i'={M.sub.P.sub.0; M.sub.P.sub.1; . . . ; M.sub.P.sub.Nc}, in which mapping codes are associated with the corresponding sets of secure mapping coefficients C.sub.N.sub.b'.sup.

(P.sub.i.sup.), C.sub.N.sub.b'.sub.-1.sup.(P.sub.i.sup.), with i=0, . . . , N.sub.c. For instance, with N.sub.b=4 quantization bits with permutations in sixteen (16) mapping codes for a permutation array P={P.sub.0, P.sub.1, . . . , P.sub.15}, results in a new permuted set of mapping codes M.sub.i'={M.sub.P.sub.0; M.sub.P.sub.1; . . . ; M.sub.P.sub.15}, which are associated with the corresponding sets of mapping coefficients given by: C.sub.5.sup.(P.sub.0.sup.)C.sub.4.sup.

(P.sub.0.sup.)C.sub.3.sup.(P.sub.0.sup.)C.sub.2.sup.(P.sub.0.sup.)C.sub.1.sup.(P.sub.0.sup.),
C.sub.5.sup.(P.sub.1.sup.)C.sub.4.sup.(P.sub.1.sup.)C.sub.3.sup.(P.sub.1.sup.)C.sub.2.sup.
(P.sub.1.sup.)C.sub.1.sup.(P.sub.1.sup.), . . . , C.sub.5.sup.(P.sub.15.sup.)C.sub.4.sup.
(P.sub.15.sup.)C.sub.3.sup.(P.sub.15.sup.)C.sub.2.sup.(P.sub.15.sup.)C.sub.1.sup.(P.sub.15.sup.).
For instance, the permutation array $P=\{7, 0, 10, 12, 6, 1, 11, 13, 4, 2, 9, 15, 5, 3, 8, 14\}$, results in the secure mapping table obtained with permutations of mapping codes as shown in Table III below.

(68) TABLE-US-00003 TABLE III Example of Mapping Table After Permutation of Mapping Codes

C.sub.5	C.sub.4	C.sub.3	C.sub.2	C.sub.1	Mapping Code	M'.sub.i
C.sub.5.sup.(p0)	C.sub.4.sup.(p0)	C.sub.3.sup.(p0)	C.sub.2.sup.(p0)	C.sub.1.sup.(p0)	M'.sub.0 = M.sub.P.sub.0	
C.sub.5.sup.(p1)	C.sub.4.sup.(p1)	C.sub.3.sup.(p1)	C.sub.2.sup.(p1)	C.sub.1.sup.(p1)	M'.sub.1 = M.sub.P.sub.1	
C.sub.5.sup.(p2)	C.sub.4.sup.(p2)	C.sub.3.sup.(p2)	C.sub.2.sup.(p2)	C.sub.1.sup.(p2)	M'.sub.2 = M.sub.P.sub.2	
C.sub.5.sup.(p3)	C.sub.4.sup.(p3)	C.sub.3.sup.(p3)	C.sub.2.sup.(p3)	C.sub.1.sup.(p3)	M'.sub.3 = M.sub.P.sub.3	
C.sub.5.sup.(p4)	C.sub.4.sup.(p4)	C.sub.3.sup.(p4)	C.sub.2.sup.(p4)	C.sub.1.sup.(p4)	M'.sub.4 = M.sub.P.sub.4	
C.sub.5.sup.(p5)	C.sub.4.sup.(p5)	C.sub.3.sup.(p5)	C.sub.2.sup.(p5)	C.sub.1.sup.(p5)	M'.sub.5 = M.sub.P.sub.5	
C.sub.5.sup.(p6)	C.sub.4.sup.(p6)	C.sub.3.sup.(p6)	C.sub.2.sup.(p6)	C.sub.1.sup.(p6)	M'.sub.6 = M.sub.P.sub.6	
C.sub.5.sup.(p7)	C.sub.4.sup.(p7)	C.sub.3.sup.(p7)	C.sub.2.sup.(p7)	C.sub.1.sup.(p7)	M'.sub.7 = M.sub.P.sub.7	
C.sub.5.sup.(p8)	C.sub.4.sup.(p8)	C.sub.3.sup.(p8)	C.sub.2.sup.(p8)	C.sub.1.sup.(p8)	M'.sub.8 = M.sub.P.sub.8	
C.sub.5.sup.(p9)	C.sub.4.sup.(p9)	C.sub.3.sup.(p9)	C.sub.2.sup.(p9)	C.sub.1.sup.(p9)	M'.sub.9 = M.sub.P.sub.9	
C.sub.5.sup.(p10)	C.sub.4.sup.(p10)	C.sub.3.sup.(p10)	C.sub.2.sup.(p10)	C.sub.1.sup.(p10)	M'.sub.10 = M.sub.P.sub.10	
C.sub.5.sup.(p11)	C.sub.4.sup.(p11)	C.sub.3.sup.(p11)	C.sub.2.sup.(p11)	C.sub.1.sup.(p11)	M'.sub.11 = M.sub.P.sub.11	
C.sub.5.sup.(p12)	C.sub.4.sup.(p12)	C.sub.3.sup.(p12)	C.sub.2.sup.(p12)	C.sub.1.sup.(p12)	M'.sub.12 = M.sub.P.sub.12	
C.sub.5.sup.(p13)	C.sub.4.sup.(p13)	C.sub.3.sup.(p13)	C.sub.2.sup.(p13)	C.sub.1.sup.(p13)	M'.sub.13 = M.sub.P.sub.13	
C.sub.5.sup.(p14)	C.sub.4.sup.(p14)	C.sub.3.sup.(p14)	C.sub.2.sup.(p14)	C.sub.1.sup.(p14)	M'.sub.14 = M.sub.P.sub.14	
C.sub.5.sup.(p15)	C.sub.4.sup.(p15)	C.sub.3.sup.(p15)	C.sub.2.sup.(p15)	C.sub.1.sup.(p15)	M'.sub.15 = M.sub.P.sub.15	

(69) The secure mapping table shown in Table III is solely for illustrative purposes and not to be limiting. The number of possible permutation arrays for sixteen (16) mapping codes is $2.0923E+13$, and the permutations of mapping codes can change dynamically over each data block or within a data block according to a predetermined pattern.

(70) In another embodiment, the secure mapper circuitry can apply a puncturing pattern to puncture mapping codes. The secure mapper circuitry can apply a puncturing array $P_u=\{p_u.sub.1, \dots, p_u.sub.N.sub.c\}$, where each $p_u.sub.i$ can represent a value of one (1) without puncturing or a value of X for puncturing, with the puncturing pattern to be applied to puncture mapping codes $M.sub.i=\{M.sub.0; M.sub.1; \dots; M.sub.N.sub.c\}$.

(71) In another embodiment, the puncturing rule is constant over the entire data block or may differ for different subsets of samples within the data block. Mapping codes can be punctured according to a predetermined rule and can be replaced by another code or a reference dummy value, and at a receiver a de-puncturing operation can be performed to recover the signal envelope. For instance, a puncturing rule $P_u=\{1, X, 1, 1, X, 1, 1, 1, 1, 1, X, 1, 1, 1, X, 1\}$, results in the punctured mapping codes $M.sub.i'=\{M.sub.0; X; M.sub.2; M.sub.3; X; M.sub.5; \dots; M.sub.9; X; M.sub.11; M.sub.12; M.sub.13; X; M.sub.15\}$, directly related to the sets of mapping coefficients:

C.sub.5.sup.(0)C.sub.4.sup.(0)C.sub.3.sup.(0)C.sub.2.sup.(0)C.sub.1.sup.(0), X, C.sub.5.sup.
(2)C.sub.4.sup.(2)C.sub.3.sup.(2)C.sub.2.sup.(2)C.sub.1.sup.(2), C.sub.5.sup.(3)C.sub.4.sup.
(3)C.sub.3.sup.(3)C.sub.2.sup.(3)C.sub.1.sup.(3), X, C.sub.5.sup.(5)C.sub.4.sup.(5)C.sub.3.sup.
(5)C.sub.2.sup.(5)C.sub.1.sup.(5), . . . , C.sub.5.sup.(9)C.sub.4.sup.(9)C.sub.3.sup.(9)C.sub.2.sup.
(9)C.sub.1.sup.(9), X, C.sub.5.sup.(11)C.sub.4.sup.(11)C.sub.3.sup.(11)C.sub.2.sup.

(11)C.sub.1.sup.(11), . . . , C.sub.5.sup.(13)C.sub.4.sup.(13)C.sub.3.sup.(13)C.sub.2.sup.

(13)C.sub.1.sup.(13), X, C.sub.5.sup.(15)C.sub.4.sup.(15)C.sub.3.sup.(15)C.sub.2.sup.

(15)C.sub.1.sup.(15).

(72) In another embodiment, the puncturing rule can be applied as a random phase rotation $C_{k,i} = e^{j\alpha} C_{k,i}$ is applied to a subset of mapping codes and corresponding mapping coefficients. The phase rotation can affect all mapping codes or subsets of mapping codes. The phase value can be predetermined and the phase rotation may change over the data block or between successive samples of the input signals' envelope according to a predetermined rule. This corresponds to a phase rotation of the values of the alphabets A_k associated with the sets of possible values of mapping coefficients.

(73) In another embodiment, the phase rotation can be applied to only a set of mapping coefficients $C_{k,i} = e^{j\alpha} C_{k,i}$, for $i = 1, \dots, n$ and $k = k_1, k_2, \dots, N'$, with $N' < N_b$.

(74) In another embodiment, a scale factor can be applied to a limited set of mapping coefficients $C_{k,i} = s_f C_{k,i}$, for $i = 1, \dots, n$ and $k = k_1, k_2, \dots, N'$, with $N' < N_b$.

(75) In FIG. 2B, based on a secure mapping table, the secure mapper circuitry **211** can generate a set of $N' \leq N_b$ active pairs of in-phase and quadrature-phase mapping components **212**- $\{1_i, \dots, N_b\}$, **212**- $\{1_q, \dots, N_b\}$ related to $N' \leq N_b$ mapping components according to the secure mapping table that can be stored in a memory element (e.g., LUT) that includes the mapping component values according to the secure mapping rule. The secure mapper circuitry **211** can generate numerical sequences with the positions of samples in which the secure mapping tables are applied or numerical sequences are previously stored in the memory elements.

(76) In another embodiment, several sets of numerical sequences having the positions of samples affected by security schemes can be stored in memory elements (e.g., LUTs).

(77) In another embodiment, the mapping table, the secure mapping table, the quantization encoding table, the quantization values table, the quantization amplitude components, or the like can be stored in memory elements (e.g., LUTs) that are operationally coupled to the quantizer circuitry, the mapper circuitry, the secure mapper circuitry, or another circuit.

(78) In another embodiment, the permutation rules, the puncturing rules, the phase rotation rules to be applied in accordance with the secure mapping tables can also be stored in memory elements (e.g., LUTs).

(79) In FIG. 2B, the secure mapper circuitry **211** can output active $N' \leq N_b$ pairs of in-phase and quadrature-phase mapping components **212**- $\{1_i, \dots, N_b\}$, **212**- $\{1_q, \dots, N_b\}$ to a set of N' bandpass modulator circuitry **213**- $\{1, \dots, N_b+1\}$, which can be configured as I/Q DACs or I/Q modulators, operable to generate the N' active bandpass components **214**- $\{1, \dots, N_b+1\}$. Without the mixer circuitry **215**- $\{O_1, \dots, O_{N_b+1}\}$, the set of N' bandpass modulator circuitry **213**- $\{1, \dots, N_b+1\}$ can generate the set of N' active bandpass components **214**- $\{1, \dots, N'\}$ with the carrier frequency f_c .

(80) In another embodiment, the secure mapper circuitry **211** can apply the phase α_n of each sample to control the initial phase of the bandpass components **214**- $\{1, \dots, N_b+1\}$ generated by the set of bandpass modulator circuitry **213**- $\{1, \dots, N_b+1\}$.

(81) In another embodiment, the set of bandpass modulator circuitry **213**- $\{1, \dots, N_b+1\}$ can generate the set of active bandpass components **214**- $\{1, \dots, N_b+1\}$ at intermediate frequency f_i and the mixer circuitry **215**- $\{O_1, \dots, O_{N_b+1}\}$ can be applied to upconvert the bandpass components **214**- $\{1, \dots, N_b+1\}$ to the carrier frequency f_c .

(82) In another embodiment, the feedback signals **219**- $\{O_1, \dots, O_{N_b+1}\}$ outputted by the set of bandpass modulator circuitry **213**- $\{1, \dots, N_b+1\}$ can be input to the

secure mapper circuitry **211** operable to generate phase and time control information for the bandpass modulator circuitry **213**-{1, . . . , N.sub.b+1} to assure time and phase synchronization of active bandpass components.

(83) In another embodiment, the feedback signals **219**-{O.sub.1, . . . , O.sub.N.sub.b.sub.+1} can be output from the set of mixer circuitry **215**-{O.sub.1, . . . , O.sub.N.sub.b.sub.+1} and input to the secure mapper circuitry **211** to generate phase and time control information to enable time and phase synchronization of active bandpass components.

(84) In another embodiment, the reference signal clock **201s-o1** having the intermediate frequency f.sub.i, can be input to the set of bandpass modulator circuitry **213**-{1, . . . , N.sub.b+1} and the input signal **201-ofc** having the carrier frequency f.sub.c can provide a reference signal to the set of mixer circuitry **215**-{O.sub.1, . . . , O.sub.N.sub.b.sub.+1} to enable up-conversion of the bandpass components **214**-{1, . . . , N.sub.b+1} to obtain carrier frequency components **216**-{1, . . . , N.sub.b+1}.

(85) In another exemplary embodiment, the output circuitry **225** can include the combiner circuitry **217**. The combiner circuitry **217** can combine the components **216**-{1, . . . , N.sub.b+1} to obtain the output signal **218**.

(86) In another exemplary embodiment, the output circuitry **225** can include a set of amplifiers (e.g., power amplifiers) with each amplifier corresponding to a certain component signal **216**-{1, . . . , N.sub.b+1}. The set of amplifiers can amplify the components **216**-{1, . . . , N.sub.b+1} to obtain a set of amplified signals. The combiner circuitry **225** can then combine the set of amplified signals to obtain the output signal **218**.

(87) FIG. 3 illustrates another embodiment of a method **300** of providing physical layer security in accordance with various aspects as described herein. In FIG. 3, the method **300** may start, for instance, at block **301** where it can include receiving samples of an input signal having information. At block **303**, the method **300** can include quantizing an amplitude of the input signal sample to obtain a set of quantization bits that represents the quantized amplitude of the input signal sample. At block **305**, the method **300** can include determining an amplitude quantization error of the input signal sample based on the difference between the amplitude of the input signal sample and the quantized amplitude of the input signal sample. At block **307**, the method **300** can include determining a compensation factor based on the amplitude quantization error. At block **309**, the method **300** can include determining a set of mapping codes or a set of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample, the compensation factor of the input signal sample, or a set of discrete amplitudes that corresponds to the set of quantization bits. At block **311**, the method **300** can include selecting a physical layer security scheme to apply to the set of mapping codes or the sets of mapping coefficients. At block **313**, the method **300** can include applying the selected security scheme to the set of mapping codes or the sets of mapping coefficients. At block **315**, the method **300** can include mapping the set of quantization bits that represents the quantized amplitude, the phase or the compensation factor of the input signal sample to obtain a set of mapping components (e.g., in-phase and quadrature-phase mapping components) based on the set of mapping codes or the corresponding sets of mapping coefficients. At block **317**, the method **300** can include modulating the secured set of mapping components to obtain a secured set of bandpass components. At block **319**, the method **300** can include upconverting the secured set of bandpass components to an intermediate frequency or a carrier frequency to obtain a secured set of bandpass components at the Intermediate Frequency (IF) or Radio Frequency (RF). At block **321**, the method **300** can include combining the secured set of bandpass components at the IF or RF frequency to obtain a secured output signal. At block **323**, the method **300** includes outputting the secured output signal that represents the secured set of bandpass components.

(88) Although the subject matter described herein may be implemented in any appropriate type of system using any suitable components, the embodiments disclosed herein may also be described in

relation to a wired or wireless network, such as the example wireless and wired networks illustrated in FIG. 4. For simplicity, the wired network portion of FIG. 4 depicts network **406** communicatively coupled to network nodes **460**, **460b**, and the wireless network portion of FIG. 4 depicts wireless devices **410**, **410b** and **410c** communicatively coupled to network nodes **460**, **460b**. In practice, a wired and wireless network may further include any additional elements suitable to support communication between wired and wired devices, wireless and wireless devices, wired and wireless devices, or between a wired or wireless device and another communication device, such as a landline telephone, a service provider, or any other network node or end device. Of the illustrated components, network node **460** and wireless device **410** are depicted with additional detail. Network **406** may provide communication and other types of services to one or more devices to facilitate the devices' access to and/or use of the services provided by, or via, the network **406**.

(89) Network **406** may comprise and/or interface with any type of communication, telecommunication, data, cellular, and/or radio network or other similar type of system. In some embodiments, network **406** may be configured to operate according to specific standards or other types of predefined rules or procedures. Thus, particular embodiments of network **406** may implement communication standards, such as Global System for Mobile Communications (GSM), Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE), Narrowband Internet of Things (NB-IoT), and/or other suitable 2G, 3G, 4G, 5G, 6G standards; wireless local area network (WLAN) standards, such as the IEEE 402.11 standards; and/or any other appropriate wireless communication standard, such as the Worldwide Interoperability for Microwave Access (WiMax), Bluetooth, Z-Wave and/or ZigBee standards.

(90) Network **406** may comprise one or more backhaul networks, core networks, internet protocol (IP) networks, public switched telephone networks (PSTNs), packet data networks, optical networks, wide-area networks (WANs), local area networks (LANs), wireless local area networks (WLANs), wired networks, wireless networks, metropolitan area networks, and other networks to enable communication between devices.

(91) Network node **460** and wireless device **410** comprise various components described in more detail below. These components work together in order to provide network node and/or wireless device functionality, such as providing wireless connections in a wireless network. In different embodiments, the wireless network may comprise any number of wired or wireless networks, network nodes, base stations, controllers, wireless devices, relay stations, and/or any other components or systems that may facilitate or participate in the communication of data and/or signals whether via wired or wireless connections.

(92) As used herein, network node refers to equipment capable, configured, arranged and/or operable to communicate directly or indirectly with a wired or wireless device and/or with other network nodes or equipment in the network to enable and/or provide access to the device and/or to perform other functions (e.g., administration) in the network. Examples of network nodes include, but are not limited to, access points (APs) (e.g., radio access points), base stations (BSs) (e.g., radio base stations, Node Bs, evolved Node Bs (eNBs), and NR NodeBs (gNBs)). Base stations may be categorized based on the amount of coverage they provide (or, stated differently, their transmit power level) and may then also be referred to as femto base stations, pico base stations, micro base stations, or macro base stations. A base station may be a relay node or a relay donor node controlling a relay. A network node may also include one or more (or all) parts of a distributed radio base station such as centralized digital units and/or remote radio units (RRUs), sometimes referred to as Remote Radio Heads (RRHs). Such remote radio units may or may not be integrated with an antenna as an antenna integrated radio. Parts of a distributed radio base station may also be referred to as nodes in a distributed antenna system (DAS). Yet further examples of network nodes include multi-standard radio (MSR) equipment such as MSR BSs, network controllers such as radio network controllers (RNCs) or base station controllers (BSCs), base transceiver stations

(BTSs), transmission points, transmission nodes, multi-cell/multicast coordination entities (MCEs), core network nodes (e.g., MSCs, MMEs), O&M nodes, OSS nodes, SON nodes, positioning nodes (e.g., E-SMLCs), and/or MDTs. As another example, a network node may be a virtual network node as described in more detail below. More generally, however, network nodes may represent any suitable wired or wireless device (or group of devices) capable, configured, arranged, and/or operable to enable and/or provide a wired or wireless device with access to the network or to provide some service to a wired or wireless device that has accessed the network.

(93) In FIG. 4, network node **460** includes processing circuitry **470**, device readable medium **480**, interface **490**, auxiliary equipment **454**, power source **486**, power circuitry **487**, antenna **462**, or the like. Although network node **460** illustrated in the example wireless network of FIG. 4 may represent a device that includes the illustrated combination of hardware components, other embodiments may comprise network nodes with different combinations of components. It is to be understood that a network node comprises any suitable combination of hardware and/or software needed to perform the tasks, features, functions and methods disclosed herein. Moreover, while the components of network node **460** are depicted as single boxes located within a larger box, or nested within multiple boxes, in practice, a network node may comprise multiple different physical components that make up a single illustrated component (e.g., device readable medium **480** may comprise multiple separate hard drives as well as multiple random access memory (RAM) modules).

(94) Similarly, network node **460** may be composed of multiple physically separate components (e.g., a NodeB component and a RNC component, or a BTS component and a BSC component, etc.), which may each have their own respective components. In certain scenarios in which network node **460** comprises multiple separate components (e.g., BTS and BSC components), one or more of the separate components may be shared among several network nodes. For example, a single RNC may control multiple NodeB's. In such a scenario, each unique NodeB and RNC pair, may in some instances be considered a single separate network node. In some embodiments, network node **460** may be configured to support multiple radio access technologies (RATs). In such embodiments, some components may be duplicated (e.g., separate device readable medium **480** for the different RATs) and some components may be reused (e.g., the same antenna **462** may be shared by the RATs). Network node **460** may also include multiple sets of the various illustrated components for different wireless technologies integrated into network node **460**, such as, for example, GSM, code division access (CDMA), wideband CDMA (WCDMA), LTE, NR, Wi-Fi, or Bluetooth wireless technologies. These wireless technologies may be integrated into the same or different chip or set of chips and other components within network node **460**.

(95) Processing circuitry **470** is configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being provided by a network node. These operations performed by processing circuitry **470** may include processing information obtained by processing circuitry **470** by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored in the network node, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination.

(96) Processing circuitry **470** may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuitry, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software and/or encoded logic operable to provide, either alone or in conjunction with other network node **460** components, such as device readable medium **480**, network node **460** functionality. For example, processing circuitry **470** may execute instructions stored in device readable medium **480** or in memory within processing circuitry **470**. Such functionality may include providing any of the various wireless features, functions, or benefits discussed herein. In some embodiments, processing circuitry **470** may include a system on

a chip (SOC).

(97) In some embodiments, processing circuitry **470** may include one or more of RF transceiver circuitry **472** and baseband circuitry **474**. In some embodiments, RF transceiver circuitry **472** and baseband circuitry **474** may be on separate chips (or sets of chips), boards, or units, such as radio units and digital units. In alternative embodiments, part or all of RF transceiver circuitry **472** and baseband circuitry **474** may be on the same chip or set of chips, boards, or units. The embodiments described by this disclosure can be implemented for the network node **460** in the interface **490** and/or the processing circuitry **470**. Further, the embodiments described by this disclosure can be implemented for the wireless device or user equipment **410** in the interface **414** and/or the processing circuitry **420**.

(98) In certain embodiments, some or all the functionality described herein as being provided by a network node, base station, eNB or other such network device may be performed by processing circuitry **470** executing instructions stored on device readable medium **480** or memory within processing circuitry **470**. In alternative embodiments, some or all of the functionality may be provided by processing circuitry **470** without executing instructions stored on a separate or discrete device readable medium, such as in a hard-wired manner. In any of those embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **470** may be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **470** alone or to other components of network node **460** but are enjoyed by network node **460** as a whole, and/or by end users and the wireless network generally.

(99) Device readable medium **480** may comprise any form of volatile or non-volatile computer readable memory including, without limitation, persistent storage, solid-state memory, remotely mounted memory, magnetic media, optical media, RAM, ROM, mass storage media (for example, a hard disk), removable storage media (for example, a flash drive, a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer-executable memory devices that store information, data, and/or instructions that may be used by processing circuitry **470**. Device readable medium **480** may store any suitable instructions, data or information, including a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **470** and, utilized by network node **460**. Device readable medium **480** may be used to store any calculations made by processing circuitry **470** and/or any data received via interface **490**. In some embodiments, processing circuitry **470** and device readable medium **480** may be considered to be integrated.

(100) Interface **490** is used in the wired or wireless communication of signaling and/or data between network node **460**, network **406**, and/or wireless devices **410**. As illustrated, interface **490** comprises port(s)/terminal(s) **494** to send and receive data, for example to and from network **406** over a wired connection. Interface **490** also includes radio front end circuitry **492** that may be coupled to, or in certain embodiments a part of, antenna **462**. Radio front end circuitry **492** comprises filters **498** and power amplifiers **496**. Radio front end circuitry **492** may be coupled to antenna **462** and processing circuitry **470**. Radio front end circuitry may be configured to condition signals communicated between antenna **462** and processing circuitry **470**. Radio front end circuitry **492** may receive digital data that is to be sent out to other network nodes or wireless devices via a wireless connection. Radio front end circuitry **492** may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **498** and/or amplifiers **496**. The radio signal may then be transmitted via antenna **462**. Similarly, when receiving data, antenna **462** may collect radio signals which are then converted into digital data by radio front end circuitry **492**. The digital data may be passed to processing circuitry **470**. In other embodiments, the interface may comprise different components and/or different combinations of components.

(101) In certain alternative embodiments, network node **460** may not include separate radio front end circuitry **492**, instead, processing circuitry **470** may comprise radio front end circuitry and may be coupled to antenna **462** without separate radio front end circuitry **492**. Similarly, in some embodiments, all or some of RF transceiver circuitry **472** may be considered a part of interface **490**. In still other embodiments, interface **490** may include one or more ports or terminals **494**, radio front end circuitry **492**, and RF transceiver circuitry **472**, as part of a radio unit (not shown), and interface **490** may communicate with baseband circuitry **474**, which is part of a digital unit (not shown).

(102) Antenna **462** may include one or more antennas, or antenna arrays, configured to send and/or receive wireless signals. Antenna **462** may be coupled to radio front end circuitry **492** and may be any type of antenna capable of transmitting and receiving data and/or signals wirelessly. In some embodiments, antenna **462** may comprise one or more omni-directional, sector or panel antennas operable to transmit/receive radio signals between, for example, 2 GHz and 6 GHz. A sub-n omni-directional antenna may be used to transmit/receive radio signals in any direction, a sector antenna may be used to transmit/receive radio signals from devices within a particular area, and a panel antenna may be a line-of-sight antenna used to transmit/receive radio signals in a relatively straight line. In some instances, the use of more than one antenna may be referred to as multiple-input multiple-output (MIMO). In certain embodiments, antenna **462** may be separate from network node **460** and may be connectable to network node **460** through an interface or port.

(103) Antenna **462**, interface **490**, and/or processing circuitry **470** may be configured to perform any receiving operations and/or certain obtaining operations described herein as being performed by a network node. Any information, data and/or signals may be received from a wireless device, another network node and/or any other network equipment. Similarly, antenna **462**, interface **490**, and/or processing circuitry **470** may be configured to perform any transmitting operations described herein as being performed by a network node. Any information, data and/or signals may be transmitted to a wireless device, another network node and/or any other network equipment.

(104) Power circuitry **487** may comprise, or be coupled to, power management circuitry and is configured to supply the components of network node **460** with power for performing the functionality described herein. Power circuitry **487** may receive power from power source **486**. Power source **486** and/or power circuitry **487** may be configured to provide power to the various components of network node **460** in a form suitable for the respective components (e.g., at a voltage and current level needed for each respective component). Power source **486** may either be included in, or external to, power circuitry **487** and/or network node **460**. For example, network node **460** may be connectable to an external power source (e.g., an electricity outlet) via an input circuitry or interface such as an electrical cable, whereby the external power source supplies power to power circuitry **487**. As a further example, power source **486** may comprise a source of power in the form of a battery or battery pack which is coupled to, or integrated in, power circuitry **487**. The battery may provide backup power should the external power source fail. Other types of power sources, such as photovoltaic devices, may also be used.

(105) Alternative embodiments of network node **460** may include additional components beyond those shown in FIG. 4 that may be responsible for providing certain aspects of the network node's functionality, including any of the functionality described herein and/or any functionality necessary to support the subject matter described herein. For example, network node **460** may include user interface equipment to allow input of information into network node **460** and to allow output of information from network node **460**. This may allow a user to perform diagnostic, maintenance, repair, and other administrative functions for network node **460**.

(106) As used herein, wireless device refers to a device capable, configured, arranged and/or operable to communicate wirelessly with network nodes and/or other wireless devices. Unless otherwise noted, the term wireless device may be used interchangeably herein with user equipment (UE). Communicating wirelessly may involve transmitting and/or receiving wireless signals using

electromagnetic waves, radio waves, infrared waves, and/or other types of signals suitable for conveying information through air. In some embodiments, a wireless device may be configured to transmit and/or receive information without direct human interaction. For instance, a wireless device may be designed to transmit information to a network on a predetermined schedule, when triggered by an internal or external event, or in response to requests from the network. Examples of a wireless device include, but are not limited to, a smart phone, a mobile phone, a cell phone, a voice over IP (VOIP) phone, a wireless local loop phone, a desktop computer, a personal digital assistant (PDA), a wireless cameras, a gaming console or device, a music storage device, a playback appliance, a wearable terminal device, a wireless endpoint, a mobile station, a tablet, a laptop, a laptop-embedded equipment (LEE), a laptop-mounted equipment (LME), a smart device, a wireless customer-premise equipment (CPE), a vehicle-mounted wireless terminal device, etc., A wireless device may support device-to-device (D2D) communication, for example by implementing a 3rd Generation Partnership Project (3GPP) standard for sidelink communication, vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-everything (V2X) and may in this case be referred to as a D2D communication device. As yet another specific example, in an Internet of Things (IoT) scenario, a wireless device may represent a machine or other device that performs monitoring and/or measurements, and transmits the results of such monitoring and/or measurements to another wireless device and/or a network node. The wireless device may in this case be a machine-to-machine (M2M) device, which may in a 3GPP context be referred to as an MTC device. As one particular example, the wireless device may be a UE implementing the 3GPP narrow band internet of things (NB-IoT) standard. Particular examples of such machines or devices are sensors, metering devices such as power meters, industrial machinery, or home or personal appliances (e.g. refrigerators, televisions, etc.) personal wearables (e.g., watches, fitness trackers, etc.). In other scenarios, a wireless device may represent a vehicle or other equipment that is capable of monitoring and/or reporting on its operational status or other functions associated with its operation. A wireless device as described above may represent the endpoint of a wireless connection, in which case the device may be referred to as a wireless terminal. Furthermore, a wireless device as described above may be mobile, in which case it may also be referred to as a mobile device or a mobile terminal.

(107) As illustrated, wireless device **410** includes antenna **411**, interface **414**, processing circuitry **420**, device readable medium **430**, user interface equipment **432**, auxiliary equipment **434**, power source **436** power circuitry **437**, or the like. Wireless device **410** may include multiple sets of one or more of the illustrated components for different wireless technologies supported by wireless device **410**, such as, for example, GSM, WCDMA, LTE, NR, Wi-Fi, WiMAX, NB-IoT, or Bluetooth wireless technologies, just to mention a few. These wireless technologies may be integrated into the same or different chips or set of chips as other components within wireless device **410**.

(108) Antenna **411** may include one or more antennas or antenna arrays, configured to send and/or receive wireless signals, and is coupled to interface **414**. In certain alternative embodiments, antenna **411** may be separate from wireless device **410** and be connectable to wireless device **410** through an interface or port. Antenna **411**, interface **414**, and/or processing circuitry **420** may be configured to perform any receiving or transmitting operations described herein as being performed by a wireless device. Any information, data and/or signals may be received from a network node and/or another wireless device. In some embodiments, radio front end circuitry and/or antenna **411** may be considered an interface.

(109) As illustrated, interface **414** comprises radio front end circuitry **412** and antenna **411**. Radio front end circuitry **412** comprises one or more filters **418** and power amplifiers **416**. Radio front end circuitry **412** is coupled to antenna **411** and processing circuitry **420** and is configured to condition signals communicated between antenna **411** and processing circuitry **420**. Radio front end circuitry **412** may be coupled to or a part of antenna **411**. In some embodiments, wireless

device **410** may not include separate radio front end circuitry **412**; rather, processing circuitry **420** may comprise radio front end circuitry and may be coupled to antenna **411**. Similarly, in some embodiments, some or all of RF transceiver circuitry **422** may be considered a part of interface **414**. Radio front end circuitry **412** may receive digital data that is to be sent out to other network nodes or wireless devices via a wireless connection. Radio front end circuitry **412** may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **418** and/or amplifiers **416**. The radio signal may then be transmitted via antenna **411**. Similarly, when receiving data, antenna **411** may collect radio signals which are then converted into digital data by radio front end circuitry **412**. The digital data may be passed to processing circuitry **420**. In other embodiments, the interface may comprise different components and/or different combinations of components.

(110) Processing circuitry **420** may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuitry, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software, and/or encoded logic operable to provide, either alone or in conjunction with other wireless device **410** components, such as device readable medium **430**, wireless device **410** functionality. Such functionality may include providing any of the various wireless features or benefits discussed herein. For example, processing circuitry **420** may execute instructions stored in device readable medium **430** or in memory within processing circuitry **420** to provide the functionality disclosed herein.

(111) As illustrated, processing circuitry **420** includes one or more of RF transceiver circuitry **422**, baseband circuitry **424**, and application processing circuitry **426**. In other embodiments, the processing circuitry may comprise different components and/or different combinations of components. In certain embodiments processing circuitry **420** of wireless device **410** may comprise a SOC. In some embodiments, RF transceiver circuitry **422**, baseband circuitry **424**, and application processing circuitry **426** may be on separate chips or sets of chips. In alternative embodiments, part or all of baseband circuitry **424** and application processing circuitry **426** may be combined into one chip or set of chips, and RF transceiver circuitry **422** may be on a separate chip or set of chips. In still alternative embodiments, part or all of RF transceiver circuitry **422** and baseband circuitry **424** may be on the same chip or set of chips, and application processing circuitry **426** may be on a separate chip or set of chips. In yet other alternative embodiments, part or all of RF transceiver circuitry **422**, baseband circuitry **424**, and application processing circuitry **426** may be combined in the same chip or set of chips. In some embodiments, RF transceiver circuitry **422** may be a part of interface **414**. RF transceiver circuitry **422** may condition RF signals for processing circuitry **420**.

(112) In certain embodiments, some or all of the functionality described herein as being performed by a wireless device may be provided by processing circuitry **420** executing instructions stored on device readable medium **430**, which in certain embodiments may be a computer-readable storage medium. In alternative embodiments, some or all of the functionality may be provided by processing circuitry **420** without executing instructions stored on a separate or discrete device readable storage medium, such as in a hard-wired manner. In any of those particular embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **420** may be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **420** alone or to other components of wireless device **410** but are enjoyed by wireless device **410** as a whole, and/or by end users and the wireless network generally.

(113) Processing circuitry **420** may be configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being performed by a wireless device. These operations, as performed by processing circuitry **420**, may include processing information obtained by processing circuitry **420** by, for example, converting the

obtained information into other information, comparing the obtained information or converted information to information stored by wireless device **410**, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination.

(114) Device readable medium **430** may be operable to store a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **420**. Device readable medium **430** may include computer memory (e.g., RAM or ROM), mass storage media (e.g., a hard disk), removable storage media (e.g., a CD or a DVD), and/or any other volatile or non-volatile, non-transitory device readable and/or computer executable memory devices that store information, data, and/or instructions that may be used by processing circuitry **420**. In some embodiments, processing circuitry **420** and device readable medium **430** may be considered to be integrated.

(115) User interface equipment **432** may provide components that allow for a human user to interact with wireless device **410**. Such interaction may be of many forms, such as visual, audial, tactile, etc. User interface equipment **432** may be operable to produce output to the user and to allow the user to provide input to wireless device **410**. The type of interaction may vary depending on the type of user interface equipment **432** installed in wireless device **410**. For example, if wireless device **410** is a smart phone, the interaction may be via a touch screen; if wireless device **410** is a smart meter, the interaction may be through a screen that provides usage (e.g., the number of gallons used) or a speaker that provides an audible alert (e.g., if smoke is detected). User interface equipment **432** may include input interfaces, devices and circuitry, and output interfaces, devices and circuitry. User interface equipment **432** is configured to allow input of information into wireless device **410** and is coupled to processing circuitry **420** to allow processing circuitry **420** to process the input information. User interface equipment **432** may include, for example, a microphone, a proximity or other sensor, keys/buttons, a touch display, one or more cameras, a USB port, or other input circuitry. User interface equipment **432** is also configured to allow output of information from wireless device **410**, and to allow processing circuitry **420** to output information from wireless device **410**. User interface equipment **432** may include, for example, a speaker, a display, vibrating circuitry, a USB port, a headphone interface, or other output circuitry. Using one or more input and output interfaces, devices and circuitry of user interface equipment **432**, wireless device **410** may communicate with end users and/or the wireless network and allow them to benefit from the functionality described herein.

(116) Auxiliary equipment **434** is operable to provide more specific functionality which may not be generally performed by wireless devices. This may comprise specialized sensors for doing measurements for various purposes, interfaces for additional types of communication such as wired communications etc. The inclusion and type of components of auxiliary equipment **434** may vary depending on the embodiment and/or scenario.

(117) Power source **436** may, in some embodiments, be in the form of a battery or battery pack. Other types of power sources, such as an external power source (e.g., an electricity outlet), photovoltaic devices or power cells, may also be used. The wireless device **410** may further comprise power circuitry **437** for delivering power from power source **436** to the various parts of wireless device **410** which need power from power source **436** to carry out any functionality described or indicated herein. Power circuitry **437** may in certain embodiments comprise power management circuitry. Power circuitry **437** may additionally or alternatively be operable to receive power from an external power source; in which case wireless device **410** may be connectable to the external power source (such as an electricity outlet) via input circuitry or an interface such as an electrical power cable. Power circuitry **437** may also be operable in certain embodiments to deliver power from an external power source to power source **436**. This may be, for example, for the charging of power source **436**. Power circuitry **437** may perform any formatting, converting, or other modification to the power from power source **436** to make the power suitable for the

respective components of wireless device **410** to which power is supplied.

(118) FIG. 5 illustrates one embodiment of an electronic device **500** in accordance with various aspects described herein. As used herein, the device **500** can be a wired device or a wireless device. Further, the device **500** may not necessarily have a user in the sense of a human user who owns and/or operates the relevant device. Instead, the device **500** may represent a device that is intended for sale to, or operation by, a human user but which may not, or which may not initially, be associated with a specific human user (e.g., a smart sprinkler controller, IoT device). Alternatively, the device **500** may represent a device that is not intended for sale to, or operation by, an end user but which may be associated with or operated for the benefit of a user (e.g., a smart power meter). The device **500** may be any device identified by the 3rd 3GPP, including a NB-IoT device, a machine type communication (MTC) device, and/or an enhanced MTC (eMTC) device. The device **500**, as illustrated in FIG. 5, is one example of a wired or wireless device such as configured for communication in accordance with one or more communication standards promulgated by the 3GPP, such as 3GPP's GSM, UMTS, LTE, 5G, 6G, Wi-Fi 6 (802.11ax), 3GPP Rel 17, DVB-S2X, wired communications, and satellite communications standards. Accordingly, the components discussed herein can be applicable to a wired or wireless device.

(119) As used herein, a wired device refers to a device that can be communicatively coupled to another device or network via a physical connection. A wired device can include a computer peripheral device, an audio or visual device, a networking device, a game console or accessory device, an industrial or medical equipment device, a home appliance device, an office device, another device, the like, or any combination thereof. A computer peripheral device can include a keyboard (e.g., coupled via USB or PS/2 cable), a mouse (e.g., coupled via USB or PS/2 cable), a monitor (e.g., coupled via HDMI, VGA, DVI, Display Port or USB cable), a printer (e.g., coupled via USB, Ethernet, or parallel port cable), a hard drive (e.g., coupled via USB, Thunderbolt or eSATA cable), a scanner (e.g., coupled via USB or Ethernet cable), a webcam (e.g., coupled via USB cable), the like, or any combination thereof. An audio or visual device can include a headphone (e.g., coupled via 3.5 mm audio jacks or USB cable), a speaker (e.g., coupled via 3.5 mm jacks, RCA cable or optical audio cable), a microphone (e.g., coupled via XLR, USB, or 3.5 mm cable), a home theater device (e.g., coupled via HDMI, RCA or optical cable), a projector (e.g., coupled via HDMI, VGA, Display Port or USB cable), the like, or any combination thereof. A networking device can include a router (e.g., coupled via Ethernet cable), a modem (e.g., coupled via Ethernet cable), a switch (e.g., coupled via Ethernet cable), an Ethernet adapter (e.g., coupled via Ethernet cable), the like, or any combination thereof. A game console or accessory device can include a game controller (e.g., coupled via USB cables), a gaming headset (e.g., coupled over USB or 3.5 mm jack cable), a console-to-TV connection (e.g., coupled over an HDMI or component cable), the like, or any combination thereof. An industrial or medical equipment device can include a CNC machine (e.g., coupled via serial or USB cable), a medical monitor (e.g., ECG, EEG), a laboratory instrument (e.g., coupled via USB or RS232 cable), the like, or any combination thereof. A home appliance device can include a wired security camera (e.g., coupled over coaxial cables), a smart thermostat (e.g., coupled over Ethernet cable), a wired smoke detector (e.g., coupled over a home's electrical system), the like, or any combination thereof. An office equipment device can include a fax machine (e.g., coupled via phone or USB cable), a wired telephone (e.g., coupled via telephone cable), a photocopier (e.g., coupled via USB or Ethernet cable), the like, or any combination thereof. Other devices can include a barcode scanner (e.g., coupled via USB or serial cable), a POS terminal (e.g., coupled via USB or serial cable), a digital camera (e.g., coupled via USB or HDMI cable), a power supply (e.g., coupled via power cable).

(120) In FIG. 5, the device **500** includes processing circuitry **501** that is operatively coupled to input/output interface **505**, RF interface **509**, network connection interface **511**, memory **515** including RAM **517**, ROM **519**, and storage medium **521** or the like, communication subsystem **531**, power source **513**, and/or any other component, or any combination thereof. Storage medium

521 includes operating system **523**, application program **525**, and data **527**. In other embodiments, storage medium **521** may include other similar types of information. Certain devices may utilize all of the components shown in FIG. 5, or only a subset of the components. The level of integration between the components may vary from one device to another device. Further, certain devices may contain multiple instances of a component, such as multiple processors, memories, transceivers, transmitters, receivers, etc.

(121) In FIG. 5, processing circuitry **501** may be configured to process computer instructions and data. Processing circuitry **501** may be configured to implement any sequential state machine operative to execute machine instructions stored as machine-readable computer programs in the memory, such as one or more hardware-implemented state machines (e.g., in discrete logic, FPGA, application specific integrated circuits (ASICs), etc.); programmable logic together with appropriate firmware; one or more stored program, general-purpose processors, such as a microprocessor or a DSP, together with appropriate software; or any combination of the above. For example, the processing circuitry **501** may include two central processing units (CPUs). Data may be information in a form suitable for use by a computer.

(122) In the depicted embodiment, input/output interface **505** may be configured to provide a communication interface to an input device, output device, or input and output device. device **500** may be configured to use an output device via input/output interface **505**. An output device may use the same type of interface port as an input device. For example, a USB port may be used to provide input to and output from device **500**. The output device may be a speaker, a sound card, a video card, a display, a monitor, a printer, an actuator, an emitter, a smartcard, another output device, or any combination thereof. device **500** may be configured to use an input device via input/output interface **505** to allow a user to capture information into device **500**. The input device may include a touch-sensitive or presence-sensitive display, a camera (e.g., a digital camera, a digital video camera, a web camera, etc.), a microphone, a sensor, a mouse, a trackball, a directional pad, a trackpad, a scroll wheel, a smartcard, and the like. The presence-sensitive display may include a capacitive or resistive touch sensor to sense input from a user. A sensor may be, for instance, an accelerometer, a gyroscope, a tilt sensor, a force sensor, a magnetometer, an optical sensor, a proximity sensor, another like sensor, or any combination thereof. For example, the input device may be an accelerometer, a magnetometer, a digital camera, a microphone, and an optical sensor.

(123) In FIG. 5, RF interface **509** may be configured to provide a communication interface to RF components such as a transmitter, a receiver, and an antenna. Network connection interface **511** may be configured to provide a communication interface to network **543a**. Network **543a** may encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network **543a** may comprise a Wi-Fi network. Network connection interface **511** may be configured to include a receiver and a transmitter interface used to communicate with one or more other devices over a communication network according to one or more communication protocols, such as Ethernet, TCP/IP, SONET, ATM, or the like. Network connection interface **511** may implement receiver and transmitter functionality appropriate to the communication network links (e.g., optical, electrical, and the like). The transmitter and receiver functions may share circuitry components, software or firmware, or alternatively may be implemented separately.

(124) RAM **517** may be configured to interface via bus **503** to processing circuitry **501** to provide storage or caching of data or computer instructions during the execution of software programs such as the operating system, application programs, and device drivers. ROM **519** may be configured to provide computer instructions or data to processing circuitry **501**. For example, ROM **519** may be configured to store invariant low-level system code or data for basic system functions such as basic input and output (I/O), startup, or reception of keystrokes from a keyboard that are stored in a non-

volatile memory. Storage medium **521** may be configured to include memory such as RAM, ROM, PROM, EPROM, EEPROM, magnetic disks, optical disks, floppy disks, hard disks, removable cartridges, or flash drives. In one example, storage medium **521** may be configured to include operating system **523**, application program **525** such as a web browser application, a widget or gadget engine or another application, and data file **527**. Storage medium **521** may store, for use by device **500**, any of a variety of various operating systems or combinations of operating systems.

(125) Storage medium **521** may be configured to include a number of physical drive units, such as redundant array of independent disks (RAID), floppy disk drive, flash memory, USB flash drive, external hard disk drive, thumb drive, pen drive, key drive, high-density digital versatile disc (HD-DVD) optical disc drive, internal hard disk drive, Blu-Ray optical disc drive, holographic digital data storage (HDDS) optical disc drive, external mini-dual in-line memory module (DIMM), synchronous dynamic random access memory (SDRAM), external micro-DIMM SDRAM, smartcard memory such as a subscriber identity module or a removable user identity (SIM/RUIM) module, other memory, or any combination thereof. Storage medium **521** may allow device **500** to access computer-executable instructions, application programs or the like, stored on transitory or non-transitory memory media, to off-load data, or to upload data. An article of manufacture, such as one utilizing a communication system may be tangibly embodied in storage medium **521**, which may comprise a device readable medium.

(126) In FIG. 5, processing circuitry **501** may be configured to communicate with network **543b** using communication subsystem **531**. Network **543a** and network **543b** may be the same network or networks or different network or networks. Communication subsystem **531** may be configured to include one or more transceivers used to communicate with network **543b**. For example, communication subsystem **531** may be configured to include one or more transceivers used to communicate with one or more remote transceivers of another device capable of wireless communication such as another wireless device, device, or base station of a radio access network (RAN) according to one or more communication protocols, such as IEEE 802.11, CDMA, WCDMA, GSM, LTE, UMTS terrestrial radio access network (UTRAN), WiMax, or the like. Each transceiver may include transmitter **533** and/or receiver **535** to implement transmitter or receiver functionality, respectively, appropriate to the RAN links (e.g., frequency allocations and the like). Further, transmitter **533** and receiver **535** of each transceiver may share circuitry components, software or firmware, or alternatively may be implemented separately.

(127) In the illustrated embodiment, the communication functions of communication subsystem **531** may include data communication, voice communication, multimedia communication, short-range communications such as Bluetooth, near-field communication, location-based communication such as the use of the global positioning system (GPS) to determine a location, another like communication function, or any combination thereof. For example, communication subsystem **531** may include cellular communication, Wi-Fi communication, Bluetooth communication, and GPS communication. Network **543b** may encompass wired and/or wireless networks such as a LAN, a WAN, a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network **543b** may be a cellular network, a Wi-Fi network, and/or a near-field network. Power source **513** may be configured to provide alternating current (AC) or direct current (DC) power to components of device **500**.

(128) The features, benefits and/or functions described herein may be implemented in one of the components of device **500** or partitioned across multiple components of device **500**. Further, the features, benefits, and/or functions described herein may be implemented in any combination of hardware, software or firmware. In one example, communication subsystem **531** may be configured to include any of the components described herein. Further, processing circuitry **501** may be configured to communicate with any of such components over bus **503**. In another example, any of such components may be represented by program instructions stored in memory

that when executed by processing circuitry 501 perform the corresponding functions described herein. In another example, the functionality of any of such components may be partitioned between processing circuitry 501 and communication subsystem 531. In another example, the non-computationally intensive functions of any of such components may be implemented in software or firmware and the computationally intensive functions may be implemented in hardware. The embodiments described by this disclosure can be implemented for the device 500 in the RF interface circuitry 509 or the transmitter circuitry 533 of the communication subsystem 531.

(129) Those skilled in the art will also appreciate that embodiments herein further include corresponding computer programs.

(130) A computer program comprises instructions which, when executed on at least one processor of an apparatus, cause the apparatus to carry out any of the respective processing described above. A computer program in this regard may comprise one or more code modules corresponding to the means or units described above.

(131) Embodiments further include a carrier containing such a computer program. This carrier may comprise one of an electronic signal, optical signal, radio signal, or computer readable storage medium.

(132) In this regard, embodiments herein also include a computer program product stored on a non-transitory computer readable (storage or recording) medium and comprising instructions that, when executed by a processor of an apparatus, cause the apparatus to perform as described above.

(133) Embodiments further include a computer program product comprising program code portions for performing the steps of any of the embodiments herein when the computer program product is executed by a computing device. This computer program product may be stored on a computer readable recording medium.

(134) Additional embodiments will now be described. At least some of these embodiments may be described as applicable in certain contexts for illustrative purposes, but the embodiments are similarly applicable in other contexts not explicitly described.

(135) In one exemplary embodiment, a method of physical layer security for codified amplification includes receiving input signal samples $s_{\text{sub},n}$ of a baseband signal

$s_{\text{sub},n} = s(t_{\text{sub},n}) = s_{\text{sub},I,n} + js_{\text{sub},Q,n}$, where $s_{\text{sub},I,n}$ and $s_{\text{sub},Q,n}$ represent the respective in-phase and quadrature-phase component of each sample $s_{\text{sub},n}$, and $t_{\text{sub},n}$ is the sampling instant. The method includes receiving clock reference signals associated with a carrier frequency $f_{\text{sub},c}$, a sampling rate $f_{\text{sub},s}$, or an intermediate frequency $f_{\text{sub},i}$. The method further includes determining, for each sample $s_{\text{sub},n}$, an amplitude

$$(136) A_n = \sqrt{s_{I,n}^2 + s_{Q,n}^2},$$

and a phase

$$(137) \theta_n = \tan^{-1} \left(\frac{s_{Q,n}}{s_{I,n}} \right)$$

of that sample $s_{\text{sub},n}$. The method further includes quantizing the amplitude $A_{\text{sub},n}$ as

$A_{\text{sub},qn} = A_{\text{sub},n} + e_{\text{sub},q}$, where $e_{\text{sub},q}$ denotes the quantization error of that sample $s_{\text{sub},n}$

based on a quantization table with $N_{\text{sub},b}$ bits, a table with the finite set of quantization values, and a table with the discrete amplitudes of quantization components $a_{\text{sub},i} = \{a_{\text{sub},1}, a_{\text{sub},2}, \dots, a_{\text{sub},N_{\text{sub},b}}\}$, with $N_{\text{sub},b'} = N_{\text{sub},b}$ or $N_{\text{sub},b'} = N_{\text{sub},b} + 1$, in which the quantized value is

decomposed, and based on an amplitude quantization error $e_{\text{sub},q}$ computing a compensation factor Δ belonging to a finite discrete alphabet. The method further includes determining a mapping table with a set of mapping codes $M_{\text{sub},i} = \{M_{\text{sub},0}; \dots; M_{\text{sub},N_{\text{sub},c}}\}$ with

$N_{\text{sub},c} = 2 \cdot \sup(N_{\text{sub},b}) - 1$, based on the magnitude $A_{\text{sub},q}$, the phase $\alpha_{\text{sub},n}$, the quantization error compensation factor Δ and the quantization table, which are associated with the sets of

mapping coefficients values $C_{\text{sub},N_{\text{sub},b},\text{sup}}(0) \dots C_{\text{sub},1,\text{sup}}(0), \dots, C_{\text{sub},N_{\text{sub},b},\text{sup}}(N_{\text{sub},c},\text{sup}) \dots C_{\text{sub},1,\text{sup}}(N_{\text{sub},c},\text{sup})$, with $N_{\text{sub},b'} = N_{\text{sub},b}$ or $N_{\text{sub},b'} = N_{\text{sub},b} + 1$, which

are defined in terms of mapping coefficients $C_{\text{sub},k} = \{C_{\text{sub},k,\text{sup}}(1); \dots; C_{\text{sub},k,\text{sup}}(N_{\text{sub},c},\text{sup})\}$.

(N.sub.c.sup.)), with $k=1, \dots, N_{\text{sub}}b'$, with the mapping coefficient values belonging to finite alphabets $A_{\text{sub}}k$. The method further includes selecting, based on the coded mapping table and the quantization table, a security scheme based on permutations of mapping codes or phase rotation or puncturing of mapping codes or scale factor multiplication to be applied in generating secure mapping table with secure mapping codes and mapping coefficients. The method further includes performing a secure mapping that applies the secure mapping table with mapping coefficients $C_{\text{sub}}k'$ where $k=1, \dots, N_{\text{sub}}b'$ with $N_{\text{sub}}b'=N_{\text{sub}}b$ or $N_{\text{sub}}b'=N_{\text{sub}}b+1$, to determine secure mapping tables with new mapping coefficients $C_{\text{sub}}k'$ to enable security on mapping the quantized value into a maximum number $N' \leq N_{\text{sub}}b$ of active pairs of in-phase $s_{\text{sub}}I,n$ and quadrature-phase $s_{\text{sub}}Q,n$ components related to $N' \leq N_{\text{sub}}b$ active mapping components given by $C_{\text{sub}}k' \cos(\alpha_{\text{sub}}n)$ and $C_{\text{sub}}k' \sin(\alpha_{\text{sub}}n)$. The method further includes mapping the quantized value into N' active mapping components and generating N' active bandpass components.

(138) In another exemplary embodiment, the method can further include a set of LUTs that include the mapping table, the secure mapping table, the security scheme, the quantization table, the sets of discrete amplitudes a_t , or the set of quantization values.

(139) In another exemplary embodiment, the maximum number of possible bandpass components and mapping coefficients $C_{\text{sub}}k'$ can be $N_{\text{sub}}b$, and the coefficients $C_{\text{sub}}k'$ can represent values from $N_{\text{sub}}b$ discrete alphabets

(140) In another exemplary embodiment, the maximum number of possible bandpass components or mapping coefficients $C_{\text{sub}}k'$ can be $N_{\text{sub}}b+1$ for $N_{\text{sub}}b$ quantization bits and coefficients $C_{\text{sub}}k'$ can represent values from $N_{\text{sub}}b+1$ discrete alphabets.

(141) In another exemplary embodiment, the bandpass modulator circuitry can be operable to receive the secure mapping coefficients to generate the bandpass components according to the secure mapping table.

(142) In another exemplary embodiment, the set of active bandpass components can be generated at a carrier frequency $f_{\text{sub}}c$ or an intermediate frequency $f_{\text{sub}}i$.

(143) In another exemplary embodiment, the method can further include generating numerical sequences having the positions of the samples to which the security schemes are applied.

(144) In another exemplary embodiment, the numerical sequences can be stored in one or more LUTs.

(145) In another exemplary embodiment, the method can further include generating pairs of in-phase and quadrature-phase mapping components and providing active pairs of mapping components to the corresponding bandpass modulator circuitry.

(146) In another exemplary embodiment, the security scheme can be based on permutations of mapping codes and corresponding mapping coefficients.

(147) In another exemplary embodiment, the security scheme can be based on phase rotations of the mapping coefficients associated with the mapping codes that can affect all or a subset of the mapping codes.

(148) In another exemplary embodiment, the security scheme can be based on a scale factor multiplication of the mapping coefficients associated with the mapping codes that can affect all or a sub-set of mapping codes; or

(149) In another exemplary embodiment, the security scheme can be based on puncturing of the mapping codes and the corresponding mapping coefficients.

(150) In another exemplary embodiment, the security scheme can include permutations, puncturing, phase rotation and scale factor multiplication of mapping codes, the like, or any combination thereof.

(151) In another exemplary embodiment, the security scheme can be applied at the same time or applied separately to different samples or subsets of samples.

(152) In another exemplary embodiment, any of the security schemes can be combined with

interleaving techniques affecting the sequence of samples of the input signal.

(153) In another exemplary embodiment, the method can further include determining control information to activate or deactivate bandpass modulator circuitry associated with the mapping components based on the secure mapping table.

(154) In another exemplary embodiment, the method can further include receiving samples $s_{\text{sub}.n}$ of a baseband signal $s(t)$ and generating in-phase $s_{\text{sub}.I,n}$ and quadrature-phase $s_{\text{sub}.Q,n}$ components of the samples $s_{\text{sub}.n}$.

(155) In another embodiment, the method can further include receiving a baseband signal and sampling, by a sampling circuitry or in a sample and hold (S/H) circuitry, the baseband signal to obtain samples of the baseband signal.

(156) In another embodiment, the bandpass modulator circuitry can be I/Q modulator circuitry or I/Q DAC circuitry.

(157) In another exemplary embodiment, the method can further include determining the amplitude $A_{\text{sub}.n}$ of the sample, quantizing the amplitude $A_{\text{sub}.n}$ to obtain the quantized amplitude $A_{\text{sub}.qn}$ of the sample, generating the mapping table, generating the secure mapping table, generating the control information, and generating the secure mapping of $N' \leq N_{\text{sub}.b}$ active pairs of in-phase and quadrature-phase components can be performed in a single method step.

(158) In one exemplary embodiment, an apparatus configured to perform physical layer security for codified amplification includes an input circuitry operable to receive samples $s_{\text{sub}.n}$ of a baseband signal $s_{\text{sub}.n} = s(t_{\text{sub}.n}) = s_{\text{sub}.I,n} + js_{\text{sub}.Q,n}$, where $s_{\text{sub}.I,n}$ and $s_{\text{sub}.Q,n}$ represent the respective in-phase and quadrature-phase components of each sample $s_{\text{sub}.n}$ and $t_{\text{sub}.n}$ is the sampling instant. The input circuitry is also operable to receive clock reference signals associated with a carrier frequency $f_{\text{sub}.c}$, a sampling rate $f_{\text{sub}.s}$, or an intermediate frequency $f_{\text{sub}.j}$. The apparatus further includes a quantizer circuitry operable to, for each sample $s_{\text{sub}.n}$, determine an amplitude

$$(159) A_n = \sqrt{s_{I,n}^2 + js_{Q,n}^2},$$

and a phase

$$(160) \theta_n = \tan^{-1} \left(\frac{s_{Q,n}}{s_{I,n}} \right)$$

of that sample $s_{\text{sub}.n}$. The quantizer circuitry is further operable to quantizing the amplitude $A_{\text{sub}.n}$ as $A_{\text{sub}.qn} = A_{\text{sub}.n} + e_{\text{sub}.q}$, where $e_{\text{sub}.q}$ denotes the quantization error of that sample $s_{\text{sub}.n}$ based on a quantization table with $N_{\text{sub}.b}$ bits, a table with the finite set of quantization values, and a table with the discrete amplitudes of quantization components $a_{\text{sub}.i} = \{a_{\text{sub}.1}, a_{\text{sub}.2}, \dots, a_{\text{sub}.N_{\text{sub}.b}}\}$, with $N_{\text{sub}.b}' = N_{\text{sub}.b}$ or $N_{\text{sub}.b}' = N_{\text{sub}.b} + 1$, in which the quantized value is decomposed, and based on an amplitude quantization error $e_{\text{sub}.q}$ computing a compensation factor Δ belonging to a finite discrete alphabet. The apparatus further includes a mapper circuitry operable to determine a mapping table having a set of mapping codes $M_{\text{sub}.i} = \{M_{\text{sub}.0}; \dots; M_{\text{sub}.N_{\text{sub}.c}}\}$ with $N_{\text{sub}.c} = 2^{\text{sup}} N_{\text{sub}.b} - 1$, based on the magnitude $A_{\text{sub}.q}$, the phase $\alpha_{\text{sub}.n}$, the quantization error compensation factor Δ and the quantization table, which are associated with the sets of mapping coefficients values $C_{\text{sub}.N_{\text{sub}.b}^{\text{sup}}}(0) \dots C_{\text{sub}.1^{\text{sup}}}(0), \dots, C_{\text{sub}.N_{\text{sub}.b}^{\text{sup}}}(N_{\text{sub}.c}^{\text{sup}}) \dots C_{\text{sub}.1^{\text{sup}}}(N_{\text{sub}.c}^{\text{sup}})$, with $N_{\text{sub}.b}' = N_{\text{sub}.b}$ Or $N_{\text{sub}.b}' = N_{\text{sub}.b} + 1$, which are defined in terms of mapping coefficients $C_{\text{sub}.k} = \{C_{\text{sub}.k}^{\text{sup}}(1); \dots; C_{\text{sub}.k}^{\text{sup}}(N_{\text{sub}.c}^{\text{sup}})\}$, with $k = 1, \dots, N_{\text{sub}.b}'$, with the mapping coefficient values belonging to finite alphabets $A_{\text{sub}.k}$. The apparatus further includes a secure mapper circuitry operable to receive the mapping table having mapping codes and mapping coefficients and selects the security scheme to apply towards determining secure mapping coefficients $C_{\text{sub}.k'}$, $k = 1, \dots, N_{\text{sub}.b}'$, and to determine secure mapping tables having new mapping coefficients $C_{\text{sub}.k'}$ required to enable security on mapping the quantized value to a maximum number $N' \leq N_{\text{sub}.b}$ of active pairs of in-phase $s_{\text{sub}.I,n}$ and quadrature-phase $s_{\text{sub}.Q,n}$ components related to $N' \leq N_{\text{sub}.b}$ active mapping components given by $C_{\text{sub}.k'} \cos(\alpha_{\text{sub}.n})$ and $C_{\text{sub}.k'} \sin(\alpha_{\text{sub}.n})$. Further, the

secure mapper circuitry is operable to determine a set of $N' \leq N_{\text{sub}.b}$ active pairs of in-phase and quadrature-phase mapping components according to the secure mapping table. In addition, the secure mapper circuitry is operable to generate numerical sequences having the positions of the samples to which the security schemes are applied in the generation of the bandpass components. The secure mapping circuitry is also operable to provide $N' \leq N_{\text{sub}.b}$ active pairs of in-phase and quadrature-phase mapping components to a set of N' bandpass modulator circuitry, which can be I/Q modulator circuitry or I/Q DAC circuitry, operable to generate the N' active bandpass components. The apparatus also includes a set of $N_{\text{sub}.b'}$ bandpass modulator circuitry with each being operable to receive the active N' pairs of in-phase and quadrature-phase mapping components and generate N' active bandpass components having the carrier frequency $f_{\text{sub}.c}$.

(161) In one exemplary embodiment, a method is performed by an electronic device. The method includes outputting an output signal that represents a set of pairs of in-phase and quadrature-phase mapping components mapped from a set of mapping codes and corresponding sets of mapping coefficients based on a set of quantization bits that represents a quantized amplitude of a sample of an input signal having information and a phase of the input signal sample. Further, the set of pairs of in-phase and quadrature-phase mapped components collectively represent the input signal sample. Each mapping code corresponds to a certain set of the sets of mapping coefficients with at least one mapping coefficient in each set of mapping coefficients being related to a set of discrete amplitudes that correspond to the set of quantization bits.

(162) In another exemplary embodiment, the method can further include quantizing the amplitude of the input signal sample to obtain the set of quantization bits that represents the quantized amplitude of the input signal sample.

(163) In another exemplary embodiment, the method can further include mapping the set of quantization bits that represents the quantized amplitude of the input signal sample and the phase of the input signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients.

(164) In another exemplary embodiment, the method can further include determining the set of mapping codes or the sets of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample, or the set of discrete amplitudes that corresponds to the set of quantization bits.

(165) In another exemplary embodiment, the method can further include selecting a physical layer security scheme to apply to the set of mapping codes or the sets of mapping coefficients; or applying the physical layer security scheme to the set of mapping codes or the sets of mapping coefficients.

(166) In another exemplary embodiment, the physical layer security scheme can include applying a scale factor or a phase rotation to the sets of mapping coefficients.

(167) In another exemplary embodiment, the method can further include determining an amplitude quantization error of the input signal sample based on the amplitude of the input signal sample and the quantized amplitude of the input signal sample; determining an amplitude quantization error compensation factor based on the amplitude quantization error; determining the set of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample and the amplitude quantization compensation factor of the input signal sample; or wherein the mapping step includes mapping the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input sample and the amplitude quantization error compensation factor of the input signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients.

(168) In another exemplary embodiment, the method can further include modulating, by a set of modulator circuitry, the set of pairs of in-phase and quadrature-phase mapped components to obtain a set of modulated signals; or combining, by a combiner circuitry, the set of modulated signals to

obtain the output signal.

(169) In one exemplary embodiment, an electronic device is operable to output an output signal that represents a set of pairs of in-phase and quadrature-phase mapped components mapped from a set of mapping codes and corresponding sets of mapping coefficients based on a set of quantization bits that represents a quantized amplitude of a sample of an input signal having information and a phase of the input signal sample. Further, the set of pairs of in-phase and quadrature-phase mapped components collectively represent the input signal sample and each mapping code corresponds to a certain set of the sets of mapping coefficients with at least one mapping coefficient in each set of secured mapping coefficients being related to a set of discrete amplitudes that correspond to the set of quantization bits.

(170) In another exemplary embodiment, the electronic device can further include a quantizer circuitry operable to quantize the amplitude of the input signal sample to obtain the set of quantization bits that represents the quantized amplitude of the input signal sample.

(171) In another exemplary embodiment, the electronic device can further include a mapper circuitry operable to map the set of quantization bits that represents the quantized amplitude of the input signal sample and the phase of the input signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients.

(172) In another exemplary embodiment, the electronic device can further include a mapping coefficient determination circuitry operable to determine the set of mapping codes or the sets of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample, or the set of discrete amplitudes that corresponds to the set of quantization bits.

(173) In another exemplary embodiment, the electronic device can further include a physical layer security scheme selection circuitry operable to select a physical layer security scheme to apply to the set of mapping codes or the sets of mapping coefficients; or a physical layer security scheme application circuitry operable to apply the physical layer security scheme to the set of mapping codes or the sets of mapping coefficients.

(174) In another exemplary embodiment, the electronic device can further include a quantization error determination circuitry operable to determine an amplitude quantization error of the input signal sample based on the amplitude of the input signal sample and the quantized amplitude of the input signal sample; a compensation factor determination circuitry operable to determine an amplitude quantization error compensation factor of the input signal sample based on the amplitude quantization error; a mapping coefficient determination circuitry operable to determine the set of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample and the amplitude quantization compensation factor of the input signal sample; or a mapper circuitry operable to map the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample and the amplitude quantization error compensation factor of the input signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients.

(175) In one exemplary embodiment, an electronic device includes a quantizer circuitry operable to quantize an amplitude of a sample of an input signal having information to obtain a set of quantization bits that represent the quantized amplitude of the input signal sample; a mapping coefficient determination circuitry operable to determine a set of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample and the phase of the input signal sample; a physical layer security scheme selection circuitry operable to select a physical layer security scheme to apply to the set of pairs of in-phase and quadrature-phase mapped components; a physical layer security scheme application circuitry operable to apply the physical layer security scheme to the set of mapping codes or the sets of mapping coefficients; a

mapper circuitry operable to map the set of quantization bits that represents the quantized amplitude of the input signal sample and the phase of the input signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients; a set of modulator circuitry operable to modulate the set of pairs of in-phase and quadrature-phase mapped components to obtain a set of modulated signals; and a combiner circuitry operable to combine the set of modulated signals to obtain an output signal.

(176) In one exemplary embodiment, a method is performed by an electronic device having a first circuitry electrically coupled to a set of amplifier circuitry operable to collectively amplify an input signal having information. The method includes outputting, by the first circuitry, a set of bandpass components that is secured by a physical layer security scheme to obtain a secured set of bandpass components. Further, the secured set of bandpass components collectively represent a sample of the input signal and correspond to both the set of amplifier circuitry and a quantized polar representation of the input signal sample.

(177) In another exemplary embodiment, each bandpass component is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample.

(178) In another exemplary embodiment, at least one amplifier circuitry is configured to output power that is linearly proportional to another of the set of amplifier circuitry or at least one amplifier circuitry is configured to output power that is non-linearly proportional to another of the set of amplifier circuitry.

(179) In another exemplary embodiment, the method includes applying the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes and obtaining the secured set of bandpass components based on the secured set of mapping codes.

(180) In another exemplary embodiment, the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

(181) In another exemplary embodiment, the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

(182) In another exemplary embodiment, the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

(183) In another exemplary embodiment, each bandpass component corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and/or a phase of the polar representation of the input signal sample.

(184) In one exemplary embodiment, an electronic device includes a set of amplifier circuitry operable to collectively amplify an input signal having information and a first circuitry electrically coupled to the set of amplifier circuitry and operable to output a set of bandpass components that is secured by a physical layer security scheme to obtain a secured set of bandpass components. Further, the secured set of bandpass components collectively represents a sample of the input signal and corresponds to both the set of amplifier circuitry and a quantized polar representation of the input signal sample.

(185) In another exemplary embodiment, the first circuitry is further operable to apply the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes and obtain the secured set of bandpass components based on the secured set of mapping codes.

(186) In one exemplary embodiment, a method performed by an electronic device having a set of

amplifier circuitry includes amplifying, by the set of amplifier circuitry, a set of output signals that collectively represents an amplified sample of an input signal having information. Further, the set of output signals corresponds to a set of bandpass components that is secured by a physical layer security scheme and is associated with a quantized polar representation of the input signal sample. (187) In another exemplary embodiment, each output signal is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the sample. (188) In another exemplary embodiment, at least one amplifier circuitry is configured to output power that is linearly proportional to another of the set of amplifier circuitry or at least one amplifier circuitry is configured to output power that is non-linearly proportional to another of the set of amplifier circuitry.

(189) In another exemplary embodiment, the method further includes applying the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes and obtaining the secured set of bandpass components based on the secured set of mapping codes.

(190) In another exemplary embodiment, each output signal corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.

(191) In one exemplary embodiment, an electronic device includes a set of amplifier circuitry operable to output a set of output signals that collectively represents an amplified sample of an input signal having information. Further, the set of output signals corresponds to a set of bandpass components that is secured by a physical layer security scheme and is associated with a quantized polar representation of the input signal sample.

(192) In another exemplary embodiment, the device further includes a first circuitry electrically coupled to the set of amplifier circuitry and operable to apply the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes and obtain the secured set of bandpass components based on the secured set of mapping codes.

(193) The previous detailed description is merely illustrative in nature and is not intended to limit the present disclosure, or the application and uses of the present disclosure. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding field of use, background, summary, or detailed description. The present disclosure provides various examples, embodiments and the like, which may be described herein in terms of functional or logical block elements. The various aspects described herein are presented as methods, devices (or apparatus), systems, or articles of manufacture that may include a number of components, elements, members, modules, nodes, peripherals, or the like. Further, these methods, devices, systems, or articles of manufacture may include or not include additional components, elements, members, modules, nodes, peripherals, or the like.

(194) Furthermore, the various aspects described herein may be implemented using standard programming or engineering techniques to produce software, firmware, hardware (e.g., circuitry), or any combination thereof to control a computing device to implement the disclosed subject matter. It will be appreciated that some embodiments may be comprised of one or more generic or specialized processors such as microprocessors, digital signal processors, customized processors and FPGAs and unique stored program instructions (including both software and firmware) that control the one or more processors to implement, in conjunction with certain non-processor circuitry, some, most, or all of the functions of the methods, devices and systems described herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more ASICs, in which each function or some combinations of certain of the functions are implemented as custom logic circuitry. Of course, a combination of the two approaches may be used. Further, it is expected that one of ordinary skill, notwithstanding

possibly significant effort and many design choices motivated by, for example, available time, current technology, and economic considerations, when guided by the concepts and principles disclosed herein will be readily capable of generating such software instructions and programs and ICs with minimal experimentation.

(195) The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computing device, carrier, or media. For example, a computer-readable medium may include: a magnetic storage device such as a hard disk, a floppy disk or a magnetic strip; an optical disk such as a CD or DVD; a smart card; and a flash memory device such as a card, stick or key drive. Additionally, it should be appreciated that a carrier wave may be employed to carry computer-readable electronic data including those used in transmitting and receiving electronic data such as electronic mail (e-mail) or in accessing a computer network such as the Internet or a LAN. Of course, a person of ordinary skill in the art will recognize many modifications may be made to this configuration without departing from the scope or spirit of the subject matter of this disclosure.

(196) Throughout the specification and the embodiments, the following terms take at least the meanings explicitly associated herein, unless the context clearly dictates otherwise. Relational terms such as “first” and “second,” and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The term “or” is intended to mean an inclusive “or” unless specified otherwise or clear from the context to be directed to an exclusive form. Further, the terms “a,” “an,” and “the” are intended to mean one or more unless specified otherwise or clear from the context to be directed to a singular form. The term “include” and its various forms are intended to mean including but not limited to. References to “one embodiment,” “an embodiment,” “example embodiment,” “various embodiments,” and other like terms indicate that the embodiments of the disclosed technology so described may include a particular function, feature, structure, or characteristic, but not every embodiment necessarily includes the particular function, feature, structure, or characteristic. Further, repeated use of the phrase “in one embodiment” does not necessarily refer to the same embodiment, although it may. The terms “substantially,” “essentially,” “approximately,” “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

Claims

1. A method, comprising: by an electronic device having a circuitry electrically coupled to a set of amplifier circuitry operable to collectively amplify an input signal having information, outputting, by the circuitry, a set of bandpass components that is secured by a physical layer security scheme to obtain a secured set of bandpass components, with the secured set of bandpass components collectively representing a sample of the input signal and corresponding to both the set of amplifier circuitry and a quantized polar representation of the input signal sample.
2. The method of claim 1, wherein each bandpass component is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample.
3. The method of claim 1, wherein at least one amplifier circuitry is configured to output power that is linearly proportional to another of the set of amplifier circuitry or at least one amplifier circuitry is configured to output power that is non-linearly proportional to another of the set of amplifier circuitry.

4. The method of claim 1, further comprising: applying the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes; and obtaining the secured set of bandpass components based on the secured set of mapping codes.
5. The method of claim 1, wherein the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.
6. The method of claim 1, wherein the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.
7. The method of claim 1, wherein the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.
8. The method of claim 1, wherein each bandpass component corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.
9. An electronic device, comprising: a set of amplifier circuitry operable to collectively amplify an input signal having information; and a circuitry electrically coupled to the set of amplifier circuitry and operable to output a set of bandpass components that is secured by a physical layer security scheme to obtain a secured set of bandpass components, with the secured set of bandpass components collectively representing a sample of the input signal and corresponding to both the set of amplifier circuitry and a quantized polar representation of the input signal sample.
10. The device of claim 9, wherein each bandpass component is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the sample.
11. The device of claim 9, wherein at least one amplifier circuitry is configured to output power that is linearly proportional to another of the set of amplifier circuitry or at least one amplifier circuitry is configured to output power that is non-linearly proportional to another of the set of amplifier circuitry.
12. The device of claim 9, wherein the first circuitry is further operable to: apply the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes; and obtain the secured set of bandpass components based on the secured set of mapping codes.
13. The device of claim 9, wherein the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.
14. The device of claim 9, wherein the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.
15. The device of claim 9, wherein the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.
16. The device of claim 9, wherein each bandpass component corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.
17. A method, comprising: by an electronic device having a set of amplifier circuitry, amplifying, by the set of amplifier circuitry, a set of output signals that collectively represents an amplified sample of an input signal having information, with the set of output signals corresponding to a set of bandpass components that is secured by a physical layer security scheme and associated with a

quantized polar representation of the input signal sample.

18. The method of claim 17, wherein each output signal is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the sample.

19. The method of claim 17, further comprising: applying the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes; and obtaining the secured set of bandpass components based on the secured set of mapping codes.

20. The method of claim 17, wherein the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

21. The method of claim 17, wherein the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

22. The method of claim 17, wherein the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

23. The method of claim 17, wherein each output signal corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.

24. An electronic device, comprising: a set of amplifier circuitry operable to output a set of output signals that collectively represents an amplified sample of an input signal having information, with the set of output signals corresponding to a set of bandpass components that is secured by a physical layer security scheme and associated with a quantized polar representation of the input signal sample.

25. The device of claim 24, wherein each output signal is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the sample.

26. The device of claim 24, wherein the device further includes a first circuitry electrically coupled to the set of amplifier circuitry and operable to: apply the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes; and obtain the secured set of bandpass components based on the secured set of mapping codes.

27. The device of claim 24, wherein the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

28. The device of claim 24, wherein the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

29. The device of claim 24, wherein the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

30. The device of claim 24, wherein each output signal corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.
